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Ministry of Education
DIRECTORATE OF CURRICULUM DEVELOPMENT

BIOLOGY
TEACHING MODULE
FORM 2
TERM 1 & 2



Developed by the Curriculum Development Centre
LUSAKA
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AUTHORS

Danny Kaoma

Deputy Headteacher, Kanjala Day Secondary School, Chipata

Himiti Munang'andu

Deputy Headteacher, Mungwi Boys Provincial STEM Secondary School, Mungwi. (ZASE)

Humphrey Gondwe

Biology Teacher, Mwembeshi Boarding Provincial STEM Secondary School, Chilanga

COORDINATORS

Martha Lisulo

Curriculum Development Specialist-Biological Sciences, Curriculum Development Centre, Lusaka

Kenneth Mudenda

Senior Curriculum Specialist –Natural Sciences, Curriculum Development Centre, Lusaka

Samson Njapau

Principal Curriculum Specialist - Natural Sciences, Curriculum Development Centre, Lusaka

Draft Zero

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VISION

Quality, life-long education for all which is accessible, inclusive and relevant to individual, national and global needs.

PREFACE

The Biology Form 2 Teaching Module has been developed to provide guidance to the teacher on how to deliver competence-based lessons that are grade-appropriate for learners at this level. The Teaching Module is aimed at providing quality education that is aligned with the Competence-Based Curriculum and 21st Century Skills. It also aims to facilitate the building of knowledge, skills, values, and positive attitudes that are intended to enable learners to live and grow into productive and useful members of their communities and society.

This Teaching Module will help the teacher lead their learners' transition from Form 1 into Form 2 of the Ordinary Secondary Education level in a systematic manner. Thus, the guidance provided in the module will transform the learner from one who merely reproduces learnt facts into a creative, critical, and an analytical thinker through selected well-thought out activities which will be done during the teaching and learning process. The suggested learning activities are designed to offer learners with hands-on and minds-on experiences through the manipulation of real objects, interaction with nature, and learning by doing. Thus, the learners will develop knowledge, desirable lifelong skills, values, and positive attitudes needed for their personal, national and global development.

The module incorporates guidance on how to deliver Competence-based learning approach. The Teaching Module will be of great use at fostering learner engagement and practical understanding of biology, rather than knowing and mere rote memorisation of biological facts. It is hoped that the guidance provided in this Teaching Module will make learning at the ordinary secondary school level more meaningful and enjoyable because it is highly activity-oriented, and allows for knowledge construction and application as a way of learning.

Kelvin Mambwe (PhD.)
Permanent Secretary- Educational Services
MINISTRY OF EDUCATION

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The contribution of the Curriculum Specialists in guiding the process and putting the pieces of work together cannot go unnoticed. Therefore, my profound gratitude goes to all the Curriculum Specialists for their hard work and commitment to duty during the development process.

Charles Ndakala (Dr.)
Director- Curriculum Development
MINISTRY OF EDUCATION

INTRODUCTION

The Biology Form 2 Teaching Module has been developed to guide teachers on how to facilitate competence-based lessons at this level. The module aims to deliver quality education that is aligned with the Competence-Based Curriculum which puts emphasis on the acquisition of the 21st Century Competences by developing prerequisite knowledge, skills, values and positive attitudes. These competences will enable learners to live and grow into productive and useful members of their communities and the Zambian society. The Teaching Module offers a guide to teachers as to how they will facilitate the learning in their biology classes through well-structured learners' activities and pedagogies. Therefore, teachers are encouraged to research widely and not only rely on the module.

At no time is the whole class approach encouraged but rather, the teacher is encouraged to facilitate learning in small cooperative groups. It is envisaged that by emphasising cooperative learning, teachers will create a more engaging and effective learning environment that will foster not only academic achievement but also essential life skills that will benefit the learners throughout their lives.

This approach aligns with the principles of constructivist learning, where learners actively construct their own knowledge through social interaction and collaborative problem-solving. In a case where a learner fails to break through, individualised learning or scaffolding should be applied so that no one is left behind.

PURPOSE OF THE MODULE

The purpose of the module is to illustrate teaching and learning activities that:

- a) are consistent with teaching and learning in a competence-based curriculum,
- b) show processes that allow achievement of 21st Century Skills such as creativity, collaboration, communication and critical thinking,
- c) demonstrate how biology can be actively taught with the teacher acting as a facilitator of the learning process,
- d) show what activity-based and learner-centred teaching and learning could look like where learners actively learn in small cooperative groups.

HOW TO USE THIS MODULE

This module provides a collection of suggested learning activities and teaching strategies designed to enhance learners' scientific competences. These activities emphasise hands-on experiences, such as manipulating real objects, interacting with nature, and learning through practice. It is crucial to understand that the desired competences and 21st-century skills will not be achieved solely by covering sub-topics but are developed through the way teaching and learning activities are carried out. Learners are expected to acquire general competences as they interact with the material under the guidance of the teacher. These essential competences include analytical thinking, effective communication, collaboration, critical thinking, environmental sustainability, real-world application, problem-solving, and the development of essential scientific skills.

The activities within this module are intended to be integrated into the teacher's existing lesson plans, not to replace them entirely. Teachers are encouraged to adapt these suggested activities to suit their specific class situation, learner abilities, and available resources. The sub-topics and activities presented in this module align with the 2024 Biology Syllabus.

Each Concept in the module begins with a “Hook” – an opening statement designed to stimulate learners' curiosity and motivate them to learn. This could be a provocative question, a relevant scenario, or something that encourages learners to think critically. Teachers are encouraged to create their own hooks that best provoke their learners' curiosity and motivation. Consider incorporating problem-solving scenarios into your hooks. In this Module, you will also find the “Content Tip”, which is intended to be for the attention of the teacher and NOT the learner, to provide guidance on the learning activities.

ICONS USED IN THIS MODULE

This module uses icons as visual symbols to represent instructions, which helps enhance the overall learning experience. By organising and categorising information, the icons make it easier for teachers to follow the content and navigate the module more efficiently.

				 Introduction
Learners' Activities	Assessment	Key Terms	Skills Developed	
			 General Competences	 Assignments
Discussion	Summary	Content Tip		

TERM 1

CONCEPT 2.1.0: CONCEPTS AND METHODS IN BIOLOGY



Introduction

Overview: Concepts and Methods in Biology encompass the fundamental ideas and techniques that explain biological phenomena used to study living organisms and their interactions. These concepts and methods form the foundation for biological research and help scientists understand the complexity of life.

Understanding this concept builds the foundation for scientific research and innovation. It equips learners to apply scientific inquiry in solving community challenges, while also fostering responsibility for conservation and environmental protection.

This concept will cover the following sub-topics: Experimental Design, Limits of Science and Scientific Ethics.



General competences:

- Analytical thinking
- Collaboration
- Communication
- Critical thinking
- Problem solving

Hook! Concepts and Methods in Biology can be compared to the magnifying glass through which we see the hidden patterns of life. Without them, Biology would just be a collection of unanswered questions.



Key Concepts/Words/Vocabulary

Sampling - choosing part of a population to represent the whole.

Methods - the strategies used to select part of a population from the whole.

Probability sampling- A technique that ensures that every member of the population has a known, non-zero chance of being selected, which minimizes bias and helps create a representative sample. Examples include; simple random sampling, systematic sampling, stratified sampling, cluster sampling and multistage sampling

Non-probability sampling- A technique that is often faster and cheaper but do not give all individuals an equal chance of being included, making the results less generalizable to the whole population. Examples include; convenience sampling, purposive sampling, quota sampling, snowball sampling and voluntary response sampling.

Independent variable- A factor that the experimenter changes or manipulates

Dependent variable- A factor that the experimenter measures or observes.

Control variables- Factors that are kept constant throughout the experiment.

Sub-Topic:2.1.1: Experimental Design



Introduction

Overview: Experimental Design is the structured process of planning how to conduct an experiment in order to obtain reliable, valid, and interpretable results. It involves carefully defining variables, selecting suitable controls, and determining how data is collected and

analysed. This Sub-Topic will allow learners to analyse sampling methods, carryout investigations using the scientific design and practise the use of two or more variables when conducting experiments.

Specific Competence: Apply experimental design in scientific investigations

1.1.1 Analysing sampling methods

Learning materials: Manila paper or flip chart, markers, bostik, slips of paper, a box, class register, coloured stickers, literature/internet source.



Activity 1: Exploring sampling methods

Procedure:

- i) Put learners into small groups of 4-5.
- ii) Ask the groups to discuss the two main types of sampling methods with their examples.
- iii) Learners should design a worksheet using manila paper and briefly fill in the worksheet such as the one below:

Worksheet

Group number.....

Names of group members:

S/N	Type of sampling method	Examples	Brief description of a sampling method
1		(i)	
		(ii)	
		(iii)	
		(iv)	
		(v)	
2		(i)	

		(ii)	
		(iii)	
		(iv)	
		(v)	

- iv) Ask each group to make presentations of their findings
- v) Make consolidations after all groups have presented.



Activity 2: Identifying sampling methods

Procedure:

- i) Put learners into small groups of 4-5
- ii) Give the groups the following scenarios;
 - A nutritionist wants to know learners' favourite snack for break time
 - An Agriculture Extension Officer wants to know a common farming method in a community
 - The School wants to know the most popular sport among Form 1 learners
- iii) Ask groups to decide which sampling method would be most suitable for each scenario given and explain why.
- iv) Ask each group to make presentations of their findings
- v) Make consolidations after all groups have presented.



Content Tip:

The two main types of sampling methods are:

1. Probability sampling (Randomised) - every member of the population has a known chance of being selected. It is divided into:
 - Simple random sampling
 - Stratified sampling

- Cluster sampling
 - Systematic sampling
2. Non-probability sampling (non-randomised) - not all members have a chance of selection; often based on convenience or judgment. It is divided into:
- Convenience sampling
 - Purposive (judgmental) sampling
 - Quota sampling
 - Snowball sampling



Activity 3: Analysing sampling methods

Procedure

- i) Put learners into small groups of 4-5
- ii) Give the groups the following scenario:

Scenario: Sampling methods in a classroom

A teacher for Physical Education (P.E) wanted to find out the favourite sport among learners in a class. She applied different sampling methods as follows:

- a) She wrote all learners' names on slips of paper, placed them in a box, and randomly picked 5 to state their favourite sport.
- b) She draws circles with names of sporting activities and asks learners to go to the sporting activity circle of their choice.
- c) She picked the first row of the class and asked them to choose their favourite sport.
- d) Using the class register, she selected every 3rd learner to mention their favourite sport.
- e) She asked the 5 learners seated in the front row to share their favourite sport.
- f) She used her own expertise and deliberately placed learners in different sporting activities

- iii) Ask groups to identify the type of sampling and explain the advantage(s) and disadvantage(s) of each method of sampling. They should tabulate the information on manila paper as shown below:

	Name of sampling method	Advantage(s)	Disadvantage(s)
a)			
b)			
c)			
d)			
e)			
f)			

- iv) Ask the groups to present their findings using the table such as the one below
v) Make consolidations after all groups have presented.

1.1.2 Carrying out investigations using the scientific design.

Learning materials: Two potted plants (any fast growing plants e.g. beans), water, cup, worksheet, manila paper/flip chart, bostik and markers.



Activity 1: Investigating steps of scientific design

Procedure:

- i) Put learners into small groups of 4-5
- ii) Ask the groups to discuss the stages of scientific design
- iii) Ask the groups to present their findings to the class
- iv) Make consolidations after all groups have presented

**Activity 2: Investigating experimental design using plant growth**

Use the same groups from Activity 1 above

Procedure:

- i) Guide the groups by posing the following question: Does the amount of light affect how fast a plant grows?
- ii) Ask the groups to predict (Learners' predictions: If a plant receives more light, then it will grow faster. If a plant receives less light then it will not grow faster)
- iii) Ask the groups to do the following:
 - a) Place one plant near a sunny window (**Plant A**).
 - b) Place the other plant in a shaded corner (**Plant B**).
 - c) Keep watering both plants equally.
- iv) Ask the groups to measure plant height every after two days with a ruler and record their observations in the table such the one below:

Plant A (sunny)	Plant B (shaded)
Day 1:cm	cm
Day 3:cm	cm
Day 5:cm	cm
Day 7:cm	cm

- v) Ask the groups to compare the growth pattern (which plant grew faster?)
- vi) Ask learners to write a short conclusion



Discussion

Learners to answer the following questions in their groups and write their answers on manila paper

1. Identify the stages of scientific design represented by steps i) to vi) in the procedure above.
 2. What other factors could affect plant growth?
- vii) Consolidate the groups answers after presentations



Content Tip:

The steps of scientific design are:

1. Identify the problem/question
2. Background research
3. Formulate a hypothesis
4. Design the experiment
5. Analyse data
6. Draw conclusions
7. Report and communicate findings
8. Refine and repeat (if necessary)

1.1.3: Practicing the use of two or more variables when conducting experiments

Learning materials: Manila paper, markers, internet source or text books, bean seedlings planted in pots, laboratory thermometer, measuring cup.



Activity 1: Exploring the variables used when conducting experiments

Procedure:

- i) Put learners into small groups of 4-5
- ii) Ask the groups to discuss the following, giving an example for each
 - a) Independent variable
 - b) Dependent variable
 - c) Control variables
- iii) Ask the groups to present their findings
- vi) Make consolidations after all groups have presented.



Activity 2: Investigating how light and water affect plant growth.

Procedure:

Use the groups from Activity 1 above

- i) Ask the groups to set up four groups of potted bean plants as follows:
 1. Less water + low light
 2. Less water + high light
 3. More water + low light
 4. More water + high light

Note: The following conditions should be observed: The plants should be

- of the same type,
- of the same size,
- planted in the same type of soil (loamy),
- planted in the same pot size and
- subjected to the same room temperature.

- ii) Ask the groups to measure plant height every after **four days** for a period of two weeks and record the observations in the table

Days	Water (ml/day)	Light (hour/day)	Plant height (cm)		
1	50	4			
4	50	8			
8	100	4			
12	100	8			

- iii) Ask groups to make bar charts using the information in the table
 iv) Consolidate the learners answers after all groups have presented



Discussion

Ask groups to identify the following from the table:

1. Independent variables
2. Dependent variable
3. Control variables

How could the experiment be improved?



Content Tip:

An independent variable;

- Is the “cause” in a cause-and-effect relationship.
- Example: The amount of sunlight given to plants, amount of water given to plants.

A dependent variable;

- Is the “effect” that responds to changes in the independent variable.
- Example: The growth of plants (height, girth of stem, number of leaves etc.).

Control variables

- Prevent outside influences from affecting the results, ensuring fairness.
- Example: Using the same type of soil, same temperature and same type of plant.



Skills Developed:

- Collaborating by playing specific assigned roles in group work and also respecting the views of other members.
- Communicating the findings through presentations of the given tasks.
- Analytical thinking through analysis of variables when conducting experiments
- Critical thinking through finding means and ways of improving experiments

Expected Standard: Experimental design applied correctly.



Assessments.

A learner wants to investigate how certain variables affects the growth of a plant. Describe an experimental design that will show and identify the following

- Independent variable.
- Dependent variable.
- Control variables.



Sub-Topic 2.1.1 Summary

- Sampling is the process of selecting a portion of a population to represent the whole.
- Proper sampling ensures results are reliable and can be generalised.
- Scientific design involves planning experiments systematically. It involves:
 1. Define the problem or question.
 2. Formulate a hypothesis.
 3. Identify variables (independent, dependent, control).
 4. Collect data through observation or measurement.
 5. Analyze results and draw conclusions.
- Using multiple variables correctly helps students understand cause-and-effect relationships

Sub-Topic 2.1.2: Limits of Science



Introduction

Overview: Science can help us understand the natural world, solve certain problems, and improve human life. However, it has boundaries or limits. It cannot answer questions of morality, spirituality, or subjective meaning. Its methods are limited to what can be tested, measured, and verified. In this sub-topic, learners will be expected to explain the limits of science such as laws, language, tools, dynamic nature of elements as well as inconsistency of elements; and classifying limits of science such as beliefs, natural laws and ecological laws.

Specific Competence: Identify the limits of science

1.2.1 Explaining the limits of science (laws, language, tools, dynamic nature of elements, inconsistency of elements)

Learning materials: Manila paper, markers, Cards or slips of paper with statements



Activity 1: Exploring and explaining the boundaries of science

Procedure:

- i) Put the learners in groups of 4-5
- ii) Ask the groups to compile a list of things that science can explain and what it cannot explain and record them in the table

S/N	What science can explain	What science cannot explain
1.		
2.		
3.		
4.		
5		
6		

- iii) Ask the groups to present their findings to the class
- iv) Consolidate the learners answers after all groups have presented



Activity 2: Explaining the limits of science

Use the same groups from Activity 1 above

Procedure:

- i) Ask Each group to discuss the limits of science (Laws, language, tools, dynamic nature of elements, inconsistencies)
- ii) Ask Groups to present their findings to the class.
- iii) Consolidate the learners' findings after all groups have presented



Discussion

Explain how your findings show that science is powerful but cannot explain everything.



Activity 3: Classifying Limits of Science

Use the same groups from Activity 1 above

Procedure:

- i) Give each group a set of statements in tabular form as shown below:
(NOTE: You can add more other statements)

Examples of statements	Classification
Prayer can heal illness	
Objects fall towards the Earth due to gravity	
Overfishing reduces fish populations and disrupts ecosystems	
The sun rises from the east and sets from the west	

Excessive cutting of trees causes accumulation of carbon dioxide in the atmosphere	
--	--

- ii) Ask the groups to classify each statement under **Beliefs**, **Natural Laws**, or **Ecological Laws**
- iii) Ask groups to present their findings to the class
- iv) Consolidate the learners answers after all groups have presented



Discussion:

Why is it important to recognise what science can and cannot explain?



Content Tip:

- Science is important for advancing knowledge, but it has clear limits in research.
- **Laws:** Science cannot break natural laws (e.g., gravity).
- **Language:** Science struggles to explain emotions or abstract ideas fully.
- **Tools:** Limited by accuracy of instruments (e.g., thermometer precision). Some research cannot be done because the necessary instruments or technology are not available.
- **Dynamic nature of elements:** Some elements change states or properties unpredictably.
- **Inconsistencies:** Results may vary due to uncontrollable factors.
- **Ethical rules** limit scientific research. Some experiments cannot be done because they may harm humans, animals, or the environment



Skills Developed:

- Collaborating by playing specific assigned roles in group work and also respecting the views of other members.
- Communicating the findings through presentations of the given tasks.
- Analytical thinking through conducting investigations.

Expected Standard: Limits of science identified accordingly



Assessments.

1. In a forest ecosystem, you notice that herbivores feed on grass, carnivores feed on herbivores, and decomposers recycle nutrients back into the soil. What type of principle is demonstrated here?
2. During a school assembly, you observe a group of students bowing their heads and closing their eyes as they begin to pray. How should this practice be classified?
3. You are walking through an orchard. Suddenly, you notice a ripe mango detach from its branch and drop straight to the ground. What principle does this event illustrate?
4. A learner claims that science can explain everything, including personal beliefs.
 - a) Explain why this statement is incorrect using one example.
 - b) Explain why language is considered a limit of science.
 - c) Give one example of how the dynamic nature of elements can limit scientific investigations.
 - d) State two reasons why inconsistencies may occur in scientific experiments.



Sub-Topic 2.1.2 Summary

- Science is powerful in explaining natural phenomena, but it has boundaries.

- Explaining limits:
 - **Laws:** Science cannot break natural or ecological laws (e.g., gravity, conservation of energy).
 - **Language:** Some human experiences, emotions, or beliefs cannot be fully expressed in scientific terms.
 - **Tools:** Scientific instruments have limits in accuracy and range.
 - **Dynamic nature of elements:** Elements and materials may change unpredictably.
 - **Inconsistencies:** Results can vary due to uncontrollable factors.
- Classifying limits:
 - **Beliefs:** Science cannot validate or invalidate matters of faith or spirituality.
 - **Natural laws:** Science must operate within fixed laws of nature.
 - **Ecological laws:** Science cannot override the balance and limits set by ecosystems.

Sub-Topic 2.1.3 Evaluate Ethics of Scientific Research



Introduction

Overview: Ethics are the principles and rules that guide what is considered right or wrong behaviour. Scientific research is guided by ethical principles that protect people, animals, and the environment. Ethics ensure that research is conducted honestly, responsibly, and with respect for human rights. In this sub-topic, learners will be expected to determine scientific ethics of a given research and apply the ethics of scientific research.

Specific Competence: Evaluate ethics of scientific research

1.3.1 Determining scientific ethics of a given research (consent, no discrimination, honesty, confidentiality)

Learning materials: Prepared case studies, Manila paper, markers, bostik.



Activity 1: Scientific ethics in research

Procedure:

- i) Put learners into small groups of 4-5
- ii) Ask each group to explain ethical principles in scientific research, focusing on consent, non-discrimination, honesty, and confidentiality.
- iii) Ask the group to present their findings
- iv) Consolidate the learners' findings after all groups have presented

**Activity 2: Ethics in action**

Use the same groups from activity 1

Procedure:

- i) Present a simple scenario: *A scientist wants to test a new medicine on a group of people.*
- ii) Ask the groups to discuss what the scientist should do to make this research ethical?
- iii) Ask the groups to present their findings
- iv) Consolidate the learners' findings after all groups have presented

**Discussion:**

Explain how these ethics protect participants and ensure trustworthy science.

**Activity 3: Case Study Role-Play****Procedure:**

- i) Put learners into small groups of 4-5

- ii) Assign each group case study examples shown below
- iii) Ask groups to identify ethical issues and propose solutions for each case study

Case Study Examples:

1. A researcher tests a new drug on people but does not inform them of possible side effects.
 - ❖ Issue:
 - ❖ Solution:
 2. A scientist excludes participants from a study based on ethnicity.
 - ❖ Issue:
 - ❖ Solution:
 3. A psychologist publishes results and reveals actual participants' names.
 - ❖ Issue:
 - ❖ Solution:
- iv) Ask groups to present their findings to the class
 - v) Consolidate the learners' findings after all groups have presented

1.3.2 Applying the ethics of scientific research

Learning materials: Short written scenarios, manila paper, markers, bostik



Activity 1: Applying ethics in scientific research

Procedure:

- i) Put learners into small groups of 4-5
- ii) Present a scenario: A scientist wants to test a new type of food supplement on students.
- iii) Ask each group what ethical principle a scientist can apply on each of the following issues:

Issue about scientific research	Ethical principle
How to ensure participants agree willingly	

How to include all students fairly	
How to report results truthfully	
How to protect participants' personal information	

- iv) Ask groups to present their findings
- v) Consolidate the learners' findings after all groups have presented



Content Tip

- The core principles of scientific ethics:
 1. **Integrity and honesty**
 - Report data truthfully without fabrication, falsification, or plagiarism.
 - Avoid manipulating results to fit expectations.
 2. **Objectivity**
 - Minimise bias in research design, data analysis, and interpretation.
 - Ensure fair peer review and publication practices.
 3. **Respect for human and animal subjects**
 - Obtain informed consent from human participants.
 - Treat animals humanely and only use them when necessary.
 4. **Social responsibility**
 - Consider the broader impact of scientific work on society and the environment.
 - Avoid harmful applications of research.
 5. **Accountability**
 - Be transparent about funding sources, conflicts of interest, and limitations.
 - Allow others to replicate and verify findings.



Skills Developed:

- Problem solving by determining scientific ethics of a given research
- Critically thinking on applying the ethics of scientific research
- Collaborating by playing specific assigned roles in group work and also respecting the views of other members.
- Communicating the findings through presentations of the given tasks.
- Analytical thinking through conducting investigations.

Expected Standard: Scientific ethics evaluated appropriately



Assessment

1. A group of Form Two students is conducting a school-based survey on study habits and access to learning resources. The survey involves students from different backgrounds, abilities, and genders. The research team must collect, analyze, and present findings while upholding ethical principles. You are part of a research team investigating how students at your school prepare for exams.
During data collection, you notice: Some students come from wealthier families and have access to private tutors, while others rely only on school resources. One respondent shares sensitive information about struggling with reading due to a learning difficulty. A team member suggests exaggerating results to make the research appear more “interesting” for presentation. Another team member proposes sharing individual responses with teachers to “help identify weak students.”
 - i. How should the team ensure that all students’ responses are treated fairly, regardless of background, ability or gender? ;
 - ii. What would be an inappropriate way to handle differences in access to resources? ;
 - iii. How should the team respond to the suggestion of exaggerating results? ;
 - iv. Why is it important to report findings truthfully, even if they seem “less exciting”? ;

- v. Should the team share individual responses with teachers? Why or why not? and
- vi. What steps can be taken to protect sensitive information provided by respondents?

2. A researcher wants to test a new type of fertilizer on farmers' crops.

- Identify one ethical principle that must be applied.
- Explain how the researcher can apply it in this case.

3. Imagine a school survey is conducted to study eating habits.

- How should confidentiality be maintained?
- Why is this important?



Sub-Topic 2.1.3 Summary

- Scientific ethics are the principles that guide researchers to conduct studies responsibly and fairly.
- Determining ethics in research involves:-
 - **Consent:** Participants must agree willingly before taking part.
 - **No discrimination:** All individuals should be treated equally regardless of background.
 - **Honesty:** Results must be reported truthfully without manipulation.
 - **Confidentiality:** Personal information of participants must be kept private.
- Applying ethics in research involves:
 - Scientists must designing and carrying out investigations that respect participants, ensure fairness, protect privacy, and maintain integrity.
 - building trust and ensuring that research benefits society without causing harm.



Concept 2.1.3 Summary

- Experimental design is the structured process of planning how to conduct an experiment in order to obtain reliable, valid, and interpretable results.
- Though science can help us understand the natural world, solve certain problems, and improve human life, it has boundaries or limits.
- Ethics of scientific research are the principles and rules that guide what is considered right or wrong when conducting a scientific research.

CONCEPT 2.2.0: PRINCIPLES OF CELLULAR LIFE



Introduction

Overview: The Principles of Cellular Life dictate that the cell is the fundamental unit of all living things, and every organism is either a single cell or a complex community of them. To sustain life, cells must perform essential functions such as respiration for energy, homeostasis for internal balance, and reproduction to pass on genetic information.

Under this concept, the following sub-topics will be covered: Micrographs, Staining Specimen and Preparation of Permanent Slides, Detailed Cell Structure and Properties of Water. While exploring these sub-topics, learners are expected to gain a range of general and Specific Competences through active participation in various learning activities.



General competences:

- Analytical thinking
- Collaboration
- Communication and
- Critical thinking.

Hook! Cells are like microscopic cities: they have power stations, factories, highways and even a city wall. Without these bustling cities, life would collapse.



Key concepts/words/vocabulary

- **Micrograph:** A photograph or digital image of an object taken through a microscope to show details invisible to the naked eye.
- **Staining:** A technique used to enhance contrast in biological samples by applying dyes that color specific cells or organelles.
- **Simple Staining:** The use of a single dye to highlight the entire microorganism so that its basic shape and structures are visible.
- **Differential Staining:** A process using multiple dyes (like the Gram stain) to distinguish between different types of organisms or cell structures based on their chemical properties.
- **Special Staining:** Targeted staining methods used to identify specific structures that are not revealed by conventional dyes, such as capsules, flagella, or spores.
- **Safranin:** A red counterstain used in cytology and histology, most commonly as the final step in Gram staining to color Gram-negative bacteria.
- **Crystal Violet:** A primary purple dye used in the Gram stain process that binds strongly to the thick peptidoglycan layer of Gram-positive bacteria.
- **Malachite Green:** A specialised green stain used primarily to detect endospores, as it can penetrate the tough spore coat with the help of heat.
- **Permanent Slide:** A specimen preserved using a mounting medium and a coverslip, designed to be stored and used repeatedly over a long period.
- **Temporary Slide:** A "wet mount" or quick preparation intended for immediate observation, usually lasting only a few hours before the specimen degrades or dries out.

- **Universal Solvent:** A substance (primarily water) capable of dissolving more substances than any other liquid, which is essential for the chemical reactions of life.
- **Surface tension:** the property of liquids' surface that allows it to resist an external force and behave like a stretched elastic membrane.
- **Capillarity:** the spontaneous flow of a liquid into a narrow tube or porous material, often moving against the force of gravity.
- **Cohesion:** the attractive force between molecules of the same substance that cause them to stick together.
- **Adhesion:** the attraction between molecules of different substances that causes them to cling to one another.
- **pH (Potential of Hydrogen):** the numerical scale used to specify the acidity or basicity of an aqueous solution.
- **SEM (Scanning electron microscope):** a high-powered imaging tool that uses a focused beam of electrons instead of light to map the surface of the sample.
- **TEM (Transmission electron microscopy):** a high-powered imaging tool that uses a focused beam of electrons instead of light to map the internal detail of the sample.
- **Light Micrographs (Photomicrographs):** Produced using visible light; these often show natural colors or artificial stains to highlight specific structures.
- **Electron Micrographs:** Use electron beams for much higher magnification and resolution.

Sub-Topic 2.3.1 Micrographs



Introduction

Overview: A micrograph is a photograph or digital image taken through a microscope to reveal more details of an object's microstructure. It provides a much higher level of enlargement to show features invisible to the naked eye. This subtopic will allow

learners to Identify types of micrographs, explain micrographs of plant and animal tissues, interpret micrographs. Furthermore, the sub-topic will enable learners communicate scientific information effectively using micrographs.

Specific Competence: Interpret micrographs.

2.1.1 Identifying types of micrographs

Learning materials: A set of micrograph images (printed or projected), Labels or cards with micrograph types, Worksheets or digital forms for recording observations.

Optional: magnified everyday objects (e.g., onion cells, insect wing) for comparison.



Activity 1: Types of micrographs

Procedure

- i) Put learners into small groups of 4 or 5.
- ii) Show learners a striking micrograph (e.g., a colorful fluorescence image of cells).
- iii) Ask learners what they notice and how they think the image was taken.
- iv) Encourage curiosity before introducing the term micrograph.
- v) Provide each group with 4–6 unlabeled micrographs (mixed types).
- vi) Ask learners to sort these images into categories based on similarities they would observe.
- vii) Ask learners to look at color, depth, detail and presence of internal structures.
- viii) Ask groups to share their sorting criteria.
- ix) Introduce the official categories (Light, Electron micrograph SEM/TEM, Fluorescence).
- x) Compare learners' observations with the provided micrographs.
- xi) Consolidate after all the groups have made presentations of their findings.

2.1.2 Explaining micrographs of plant and animal tissues (Plant: palisade, spongy, epidermal; animal: blood cells, muscle cells, nerve cells)

Learning materials: Sets of micrograph images (printed cards or projected slides), Worksheets with guiding questions, markers of different colours/sticky notes for group work.

Optional: microscopes with prepared slides (if available).



Activity 1: Explaining micrographs of plant and animal tissues

Procedure

- i) Put learners into small groups of 4-5
- ii) Provide each group with a mixed set of plant and animal tissue micrographs.
- iii) Ask learners to sort these images into plant versus animal tissues.
- iv) Ask learners to name the specific tissue type.
- v) Ask groups to share their sorting criteria.
- vi) Consolidate after all the groups have made presentations of their findings.



Content Tip:

- Encourage learners to look for clues such as cell shape, arrangement, presence of organelles, colour/staining.
- **Plant Tissues:**
 - **Palisade Mesophyll:** Column-shaped, tightly packed cells near the leaf surface, maximised for light absorption through numerous chloroplasts.
 - **Spongy Mesophyll:** Loosely arranged with large air spaces to facilitate gas exchange within the leaf.

- **Epidermis:** The outermost protective layer, often covered by a waxy cuticle to prevent water loss.
- **Animal Tissues:**
 - **Blood Cells:** Red blood cells appear as biconcave disks without nuclei, used to transport oxygen; white blood cells are larger with distinct, often lobed nuclei in phagocytes for engulfing foreign body; and large nucleus in Lymphocytes.
 - **Muscle Cells:** Long, fiber-like cells; skeletal and cardiac types show "striations" (stripes) while smooth show spindle shape for contraction.
 - **Nerve Cells (Neurons):** Characterized by a cell body with long, branching extensions (axons and dendrites) to transmit signals.

2.1.3 Interpreting micrographs

Learning materials: A set of unlabeled micrographs (plant tissues: palisade, spongy, epidermal; animal tissues: blood cells, muscle cells, nerve cells), Guiding question cards, Worksheets or digital forms for recording observations.

Optional: microscopes with prepared slides for hands-on comparison.



Activity 1: Interpreting micrographs

Procedure

- i) Put learners into small groups of 4-5.
- ii) Show the learners a sample micrograph.
- iii) Ask learners to relate the structure of the cell in the micrograph to its function.
- iv) Give each group assorted micrographs of plant and animal cells.
- v) Ask the groups to examine each image carefully.

vi) Ask the groups to write down what they notice (shape, arrangement, presence of organelles, spaces, staining) in the following table.

Visible Feature(s)	Organelle Name	Primary Function

vii) Ask groups to share their interpretations.

viii) Consolidate the findings after all the groups have presented.



Content Tip:

- Interpretation involves analysing an organelle's size, shape, and position:
- Size: The nucleus is typically the largest and most prominent organelle.
- Contrast: Stains like iodine (for plants) or methylene blue (for animals) are used to make clear structures visible under light microscopes.
- Unique structures: Presence of a cell wall or large vacuole immediately identifies a specimen as plant-based.

2.1.4 Communicating scientific information effectively using micrographs

Learning materials: Sets of micrographs (plant: palisade, spongy, epidermal; animal: blood cells, muscle cells, nerve cells), manila paper or digital slides for presentations, markers of assorted colours/sticky notes for group work.



Activity 1: Communicating scientific information effectively using micrographs

Procedure

- i) Put learners into small groups of 4-5.
- ii) Give each group assorted micrographs of plant and animal cells/tissues.
- iii) Ask groups to observe the micrograph carefully.
- iv) Ask groups to write down what they see in terms of structure, arrangement, colour/staining.
- v) Ask groups to relate the cell/tissue type and its function.
- vi) Ask groups make presentations to the class.
- vii) Collaboratively consolidate the learner's presentation.



Skills Developed:

- Collaborating by playing specific assigned roles in group work and also respecting the views of other group members.
- Communicating the findings through presentations of the given tasks.
- Analytical thinking through interpretation of micrographs.
- Critical thinking through explanation of micrographs of plant and animal cells/tissues.

Expected Standards: Micrographs interpreted correctly



Assessment

1. Explain the primary difference between a **photomicrograph** and an **electron micrograph**. In your answer, discuss why a scientist might choose an electron microscope over a light microscope when studying a virus.
2. Why is it essential for every micrograph to include a **scale bar**? Describe what could happen if a medical researcher shared an image of a bacteria cell without providing any measurement reference.
3. Imagine you are looking at a micrograph of a plant cell and an animal cell. Describe the specific microstructures you would expect to see in the plant cell micrograph that would allow you to tell them apart.



Sub-Topic 2.2.1 Summary

- A micrograph is a photograph or digital image taken through a microscope to capture features of a specimen that are invisible to the naked eye.
- They are categorised by the technology used, such as photomicrographs (light microscopes) and electron micrographs (Scanning or Transmission Electron Microscopes).
- They provide essential data for morphological characterisation, allowing scientists to study the size, shape, and arrangement of cells, tissues, or material structures.
- Every micrograph must include a scale bar to provide a reference for the actual size of the objects, which is critical for quantitative analysis.

Sub-Topic 2.2.2: Staining Specimen and Preparation of Permanent Slides



Introduction:

Overview: Staining is a technique used to enhance contrast in biological samples by applying dyes that color specific cells or organelles. A permanent slide is a specimen preserved using a mounting medium and a coverslip, designed to be stored and used repeatedly over a long period.

This Sub-topic will allow learners to explore staining techniques on specimen, staining of specimen. Additionally, the topic will equip learners with skills on how to prepare permanent plant tissue slides.

Specific Competence: Apply appropriate staining techniques on specimen.

2.2.1: Exploring staining techniques on specimen.

Learning materials: Textbooks/handouts, Internet connectivity and electronic gadgets, white candle, plain white papers, powdered charcoal



Activity 1: staining at work.

Procedure

- i) Put learners into small groups of 4-5.
- ii) Provide papers with written letters using white candle.
- iii) Ask learners to identify the letters.
- iv) Ask learners to sprinkle charcoal powder on the papers and rub gently with hands.
- v) Ask learners to repeat procedure (iii).



Discussion:

- Compare the visibility of the letters before and after applying charcoal powder and give explanations for the difference.



Activity 2: Exploring staining techniques on specimen (Simple staining)

Procedure

- Use the same groups in activity 1
- Ask learners to read some literature on simple staining.
- Ask learners: “Why do we need stains when using a microscope?”
- Ask the learners to answer the following research questions
 - Describe a simple stain?
 - List three common dyes used in simple stain and their colours.
 - When is a simple stain preferred over others?
- Ask groups to make presentations to the class.
- Collaboratively consolidate the learners’ presentation.



Content Tip:

- The common dyes used in simple staining include safranin (stains cells pink or red), methylene blue (stains cells blue) and crystal violet (stains cells purple)
- If dyes are not readily available, ask learners to research through videos on internet and then make presentations; consolidate their findings.



Activity 3: Exploring staining techniques on specimen (Differential Staining)

Procedure

- i) Put learners into small groups of 4-5.
- ii) Ask learners to read some literature on differential staining.
- iii) Ask learners to answer the following questions.
 - Describe differential staining?
 - Compare Gram stain and Acid-fast stain based on the features listed in the table. (Record comparisons in the following table).

Feature	Gram Stain	Acid-Fast Stain
Target cell wall component		
Primary stain		
Role of the mordant		
Decolorising agent		
Counterstain		
Colour of positive result		
Colour of negative result		
Common example organism		

- Why are mordants and counterstains used?
 - Give one example for each. (mordant and counterstain)
- iv) Ask groups to make presentations to the class
 - v) Instruct the audience (other groups) to take notes during presentations.
 - vi) Collaboratively consolidate the learner's presentation.



Content Tip:

- Key examples of differential staining include Gram staining-used to classify bacteria into two major groups based on the characteristics of their cell walls;
- and acid-fast stain-used to identify bacteria with waxy, lipid-rich cell walls.

2.2.2: Staining of Specimen

Learning materials: Prepared slides or fresh specimens (e.g., onion epidermis, bacterial cultures), stains such as methylene blue, crystal violet, safranin, Gram's iodine or malachite green, microscope, microscope glass slides and cover slips, droppers, distilled water, alcohol (for decolorisation), notebook and pencils.



Activity 1: Simple staining

Procedure

- Put learners into small groups of 4-5.
- Make all the necessary requirements available for each group
- Ask learners to place a drop of water on a clean glass slide
- Ask learners to carefully peel a thin, transparent layer from the inside of an onion scale leaf using forceps.
- Ask learners to place the epidermal layer on a glass slide.
- Ask learners to stain onion epidermis with iodine/ methylene blue/ safranin.
- Ask learners to allow the stain to sit for 1-2 minutes so that the cells absorb the dye.
- Ask learners to place a cover slip over the specimen, avoiding air bubbles.
- Ask learners to observe under microscope.
- Ask learners to sketch their observations.
- Ask each group to share their findings with the class

xii) Consolidate the learners' findings.



Content Tip:

- In situations where microscopes and dyes are not readily available, ask learners to research through videos on internet and then make presentations; consolidate their findings.



Activity 2: Differential staining (Gram stain)

Procedure

- Put learners into small groups of 4-5.
- Make all the necessary requirements available for each group
- Ask learners to place a drop of sterile (bacteria-free) water on a clean slide.
- Ask learners to carefully transfer a small amount of bacterial colony on to a glass slide. (Avoid contamination).
- Ask learners to spread the specimen into a thin layer to form an even smear.
- Ask learners to heat fix by passing the glass slide through a flame two to three times.
- Ask learners to flood the smear with crystal violet.
- Ask learners to wait 30 to 60 seconds, then rinse gently with water.
- Ask learners to flood the smear with iodine.
- Ask learners to wait 60 seconds, then rinse gently with water
- Ask learners to tilt the slide.
- Ask learners to apply 95% ethanol or acetone-alcohol drop by drop until runoff runs clear (usually 5–10 seconds)
- Ask learners to rinse immediately with water.
- Ask learners to flood the smear with safranin.

- xv) Ask learners to wait 30–60 seconds then rinse with water.
- xvi) Ask learners to dry the glass slide with blotting paper (do not rub)
- xvii) Ask learners to observe under microscope.
- xviii) Ask learners to sketch their observations.
- xix) Ask each group to share their findings with the class
- xx) Consolidate the learners' findings.



Content Tip:

A tour to the hospital/ clinic laboratory would be preferred. Alternatively, videos from the internet would be researched to enhance understanding.

2.2.3: Preparing permanent plant tissue slides

Learning materials: Fresh plant stems (e.g., sunflower), sharp razor blades or a microtome or scalpel, glass slides, and coverslips, microscope, 70% ethanol, safranin (for lignified tissues like xylem) and Fast Green (for cellulose), Graded alcohol series (50%, 70%, 95%, 100%) and Xylene (clearing agent), Canada Balsam or DPX resin.



Activity 1: Preparing permanent plant tissue slides.

Procedure

Day 1

- i) Put learners into small groups of 4-5.
- ii) Make all the necessary requirements available to the learners.
- iii) Ask learners to cut extremely thin transverse sections of the plant material using a razor blade.

- iv) Ask learners to place cut sections immediately into a watch glass with water to prevent drying.
- v) Ask learners to submerge the best specimen of the cut sections in 70% Ethanol for at least 24 hours to preserve cell structure and stop metabolic activity.

Day 2

- vi) Ask learners to place cut sections in Safranin for 2–5 minutes.
- vii) Ask learners to wash off the safranin with water, then quickly dip in Fast Green (approximately 10–30 seconds) to create a double-stain effect.
- viii) Ask learners to move the stained sections through a series of increasing alcohol concentrations (50% → 70% → 90% → 100%) for 5–10 minutes each. This removes all water from the cut sections, which is essential for resin mounting.
- ix) Ask learners to immerse the tissue in Xylene or clove oil until it becomes transparent.
- x) Ask learners to place the section on a clean slide and add a small drop of Canada Balsam or DPX.
- xi) Ask learners to gently lower the coverslip at a 45° angle to avoid trapping air bubbles.
- xii) Ask learners to allow the slide to air-dry or use a warming plate at 60°C to harden the resin and label with the plant name, tissue type, and date.
- xiii) Ask learners to view the slide under low and high power.
- xiv) Ask learners to identify xylem (stained red/pink) and phloem/parenchyma (stained green).
- xv) Ask learners to sketch a labeled diagram of the tissue arrangement, noting the clarity and quality of the staining.
- xvi) Make a gallery of the drawings from the groups.
- xvii) Ask groups to walk around the gallery and compare the drawings.
- xviii) Consolidate the learners' findings.



Discussion:

- Why is dehydration critical when preparing permanent slides?



Content Tip:

- Alternatively, videos from the internet would be researched and then learners present their findings. Consolidate after all groups have presented to enhance understanding.
- Permanent slide preparation begins with fixation to stabilise cell structures, followed by dehydration and clearing to remove water.
- The specimen is embedded in wax for thin slicing and stained with dyes to create visual contrast.
- Finally, it is mounted under a coverslip to ensure long-term preservation and stability for study.



Skills Developed:

- Collaborating by playing specific assigned roles in group work and also respecting the views of other group members.
- Communicating the findings through presentations of the given tasks.
- Analytical thinking during staining and observation of specimen.
- Critical thinking through explaining staining techniques on specimen.

Expected Standard:

Appropriate staining techniques on specimen applied accordingly.



Assessment:

1. A virologist is investigating a newly discovered virus. She first tries to observe it using a light microscope but struggles to see any clear details. Later, she switches to an electron microscope and obtains a much sharper image.

Based on this scenario;

- i. Explain the primary difference between a photomicrograph and an electron micrograph,
 - ii. Why was the electron microscope the better choice for studying the virus?
2. A medical researcher publishes an image of a bacterial cell in a journal article. However, the image does not include a scale bar. Another scientist misinterprets the size of the cell, leading to confusion about the bacterium's potential impact on human health.

In this scenario;

- i. Why is it essential for every micrograph to include a scale bar?
- ii. What problems could arise when images are shared without any measurement reference?



Sub-Topic 2.2.2 Summary

- Staining is used to increase the contrast of naturally transparent biological specimens, making specific cell structures (like the nucleus or cell wall) clearly visible under a microscope.
- The types of Staining Techniques include: simple staining (uses a single dye e.g., Methylene Blue to show basic shape), differential staining (uses multiple dyes to distinguish between different types of cells or structures).
- Before staining, specimens must be treated using heat or chemicals (like formalin) to prevent decay and "lock" the internal architecture in place.
- To prepare a permanent plant slide, extremely thin transverse sections of a stem or root are cut using a sharp razor to ensure light can pass through the sample.

- For plants, sections are often stained with Safranin (red) and then passed through a series of increasing alcohol concentrations (30% to 100%) to remove all water.
- The dehydrated and stained section is placed in a resinous medium like DPX or Canada Balsam and sealed under a coverslip to protect it from environmental damage for long-term study.

Sub-Topic 2.2.3: Detailed Cell Structure



Introduction:

Overview: Detailed cell structure is important to help learners to have a deeper understanding of the structure of cells and their functions. Learners will explore different cell organelles and how each one contributes to the overall functioning of the cell through observation, discussion and practical activities. This sub-topic will focus on drawing a detailed cell structure, designing detailed models of cells and applying knowledge of organelles and their functions in order to understand cellular functions.

Specific Competence: Illustrate detailed cell structure of an organism

2.3.1 Drawing a detailed cell structure

Learning materials: Prepared slides/permanent slides, Plain A4 papers, bostik, pencils,



Activity 1: Observing and sketching cells

Procedure:

- Put learners into small groups of 4-5

- ii) Make available prepared slides/ permanent slides of plant and animal cells.
- iii) Place microscopes at different stations.
- iv) Allow the groups to use the microscopes to observe the given prepared slides.
- v) Ask learners to sketch a quick rough outline of the cell and label the visible parts.
- vi) Ask learners to create a gallery of group drawings and allow learners to take a walk through the gallery and compare the drawings.
- vii) Ask learners to discuss their challenges during the activity and share with the class.
- viii) Consolidate presentations from the groups.

2.3.2 Designing detailed models of cells

Learning materials: Base cardboard, play dough/ clay soil, small pipes/sticks, beads, label stickers, cutout organelles.



Activity 1: Designing a model of a detailed cell

Procedure:

- i) Put learners into small groups of 4-5
- ii) Display a well labelled diagram of a detailed cell consolidated from learning activity 2.3.1 to show where the organelles are found.
- iii) Ask each group to create a 2D/3D model on the provided base cardboards.
- iv) Ask learners to place the cutout papers for organelles in the model to show positions of the organelles.
- v) Ask learners to make a gallery of models and compare their work with others.
- vi) Award marks for the models satisfying the standard.



Activity 2: Cell walk

Procedure:

- i) At the central part of the classroom or ground, ask learners to make a sketch diagram of a detailed cell/ make a big model of a detailed cell of about 2 metres on a cardboard.
- ii) Ask learners to place labels for each organelle on the sketch diagram or model
- iii) Using the groups in activity 1, ask one member to walk through the sketch of the cell and pick a label for an organelle.
- iv) In groups, ask learners to discuss the functions of the cell organelles.
- v) Ask groups to make presentations.
- vi) Consolidate the learners' findings after all groups have presented

2.3.3 Applying knowledge of organelles and their functions in order to understand cellular functions

Learning materials: index cards (cell organelles, functions and cellular processes), manila paper, markers, bostik



Activity 1: Function-fit challenge

Procedure:

- i) Put learners into small groups of 4-5.
- ii) Ask groups to pick the already made index cards containing names of cell organelles and names of cellular processes.
- iii) Ask learners to complete the table by matching the index cards.

Cell organelle	Function	Cellular process its involved in

- iv) Ask learners to make presentations.
- v) Consolidate the learners' findings after all groups have presented



Content Tip:

- The organelles to focus on are; mitochondria, ribosomes, chloroplasts, Endoplasmic reticulum (ER), Golgi bodies, nucleus and vacuole.
- Cellular processes include respiration, photosynthesis, cell division and protein synthesis.



Skills Developed:

- Collaboration through group work.
- Communicating the findings through presentations of given tasks.
- Analysing cell structure in relation to function.

Expected Standard- Detailed cell structure of an organism illustrated accordingly



Assessment:

1. Discuss how cell organelles interact to carry out cellular functions.
2. A cell is observed to have many lysosomes. Predict what type of cell this might be and explain why.
3. Explain how the mitochondrion and chloroplasts work together in plant cells to maintain energy balance.
4. Design a 3D model of a plant cell.



Sub-Topic 2.2.3 Summary

- A detailed cell structure whether plant or animal should have cell organelles that enable cells conduct different cell processes.

- Cell organelles include; mitochondria, ribosomes, endoplasmic reticulum, Golgi body and nucleus
- Life processes supported by cell organelles include respiration, cell division, nutrition, control and coordination, storage of materials, transport etc.

Sub-Topic 2.2.4 Properties of water



Introduction:

Overview: Water is essential for all living systems. Its unique properties enable biochemical reactions, stabilise environments for cells, transport substances and support the structure and energetics of life at the cellular level. In this sub-topic, learners will investigate the properties of water in relation to supporting life and also evaluate the importance of water in a cell.

Specific Competence: Analyse the properties of water in relation to supporting life

2.3.1 Investigating properties of water in relation to supporting life (*transparent to light, universal solvent, liquid at room temperature, has neutral pH, surface tension, its density reduces when in solid form, ability to resist temperature changes*)

Learning materials: Distilled/tap water, potted plants, oil, salt, sugar, graduated cylinders, clear cups, scoops/spoons, handouts/online search data, pH universal indicator paper/solution, food colour/potassium permanganate, ice cubes, thermometer, paper clip/small floating objects, dropper/pipette, Torch light.



Activity 1: Exploring water properties

Procedure

- Put learners into small groups of 4-5.
- Ask learners to research the properties of water.
- Ask learners to explain each of the properties of water identified.
- Ask learners to make group presentations.

- v) Consolidate the findings after the presentation.



Activity 2: Investigating properties of water - water as a universal solvent

Procedure

- Put learners into small groups of 4-5
- Ask each group to collect clear cups, the same amount of water, salt, sugar and oil.
- In separate cups, ask learners to mix equal amounts of each solute with water
- Ask learners to write the observation of the solubility of the given substance in the table below.

Name of substance	State whether it dissolves in water

- Ask groups to make presentations
- Consolidate after presentations from the groups



Discussion:

- Explain why some substances are able to dissolve in water while others do not.
- Relate water's property of dissolving many substances to supporting life.



Activity 3: Investigating properties of water - water is transparent

Procedure:

- Put learners into small groups of 4-5.

- ii) Ask learners to fill a clear beaker with water
- iii) Using a torch light ask learners to shine the light through the beaker containing the clear water.
- iv) Ask learners to add a drop of food colouring to the water.
- v) Using a torch light ask learners to shine the light through the beaker containing the coloured water.
- vi) Ask learners to compare the results and draw conclusions.
- vii) Ask learners to make presentations.
- viii) Consolidate the group findings after presentations



Discussion:

- Discuss how the property of water being transparent supports life.



Activity 4: Investigating properties of water - water is liquid at room temperature

Procedure:

- i) Put learners into small groups of 4-5.
- ii) Ask learners to record the temperature of the room.
- iii) Ask learners to put water in a beaker.
- iv) Ask learners to record the physical state of water.
- v) Ask the learners to record the temperature of water.
- vi) Ask learners to compare the temperature of the water with that of the room.
- vii) Ask groups to make presentations.
- viii) Consolidate learners' findings after the presentations are done.



Discussion:

- Discuss the benefit of water being liquid at room temperature supports life.



Activity 5: Investigating properties of water - water's reduction in density when in solid form

Procedure:

- Put learners into small groups of 4-5
- Ask learners to place ice cubes in water and observe
- Ask the groups to present their findings
- Consolidate the learners' findings



Discussion:

- Explain how the observations made make water beneficial to supporting life



Activity 6: Investigating properties of water - water is resistant to temperature changes

Procedure:

- Put learners into small groups of 4-5
- Ask learners to put equal amounts of water and oil in two separate beakers
- Ask learners to heat water and oil in the beakers and compare temperature rise of water and oil for a period of 5 minutes.

Time	start	1min	2min	3min	4min	5min
Water temperature						
Oil temperature						
Difference						

- iv) Ask the groups to present their findings
- v) Consolidate the learners' findings



Discussion:

- Discuss how the observations made helps water to support life.



Activity 7: Investigating properties of water - water has a neutral pH

Procedure:

- i) Put learners into small groups of 4-5.
- ii) Ask learners to put water in a beaker.
- iii) Ask learners to test water with universal indicator paper
- iv) Ask learners to record the pH of water.



Discussion:

- Relate water's neutral pH to supporting of life.



Activity 8: Investigating properties of water - surface tension of water

Procedure:

- i) Put learners into small groups of 4-5.
- ii) Ask learners to put water in a beaker.
- iii) Ask learners to place a paper clip/small floating object gently on the surface of water and observe.
- iv) Ask groups to make presentations of their observations.
- v) Consolidate the findings after the presentations



Discussion:

- Discuss how the surface tension of water supports life.



Content Tip:

- Ice is less dense than liquid water.
- The properties of water are essential making it:
 - i) Vital for transporting nutrients in organisms.
 - ii) Prevent bodies of water from freezing solid and protecting aquatic life.
 - iii) Helpful for organisms to maintain stable internal conditions.
 - iv) Act as a medium for biochemical reactions.
 - v) Provide a stable environment for cellular processes.
 - vi) Allow small organisms (like insects) to move across water surfaces.
 - vii) Allowing aquatic plants to conduct photosynthesis and aquatic organisms to receive light.

2.3.2 Evaluating importance of water in a cell (*medium of chemical reaction, acts as a solvent, turgidity in plant cells*)

Learning materials: sets of cards stating properties of water, sets of cards showing cellular implications, cardboard, glue.



Activity 1: water property card sorting

Procedure:

- i) Put learners into small groups of 4-5.
- ii) Ask the learners to sort the cards of water properties and cellular implications and pair them.
- iii) Ask learners to stick the paired cards on the cardboard with glue.
- iv) Ask learners to make a gallery of the group presentations
- v) Consolidate the findings after the presentations.



Skills Developed:

- Collaboration through group work.
- Communicating the findings through presentations of given tasks.
- Evaluating the properties of water
- Analysing properties of water

Expected Standard- Properties of water in relation to supporting life analysed accordingly



Assessment

The secret life of a pond. One morning as you are walking along a shallow pond, you notice the sun rays striking and penetrating through the pond. The water slowly warms up. Since the surface of the water was unusually calm, you notice several insects effortlessly skating across while their feet created tiny dents on the surface. Taking a closer look, you notice some mosquito eggs floating and held together at the surface. Answer the following questions basing on the scenario given

1. What property of water allows the insects to walk on the surface of water?
2. Why is the specific heat capacity of water important for the survival of aquatic life?
3. How does light rays penetrating through the water in the pond benefit aquatic plants?



Sub-Topic 2.2.4 Summary

- Water is essential to life
- Water is a solvent, transparent, has a neutral PH,
- Water has adhesion and cohesion forces.
- Water is a liquid at room temperature and pressure
- Water plays a major role in the structure, metabolism, transport and temperature regulation due to its unique physical and chemical properties.



Concept 2.2.0 Summary

- The Principles of Cellular Life dictate that the cell is the fundamental unit of all living things.
- A micrograph is a photograph or digital image taken through a microscope to reveal more details of an object's microstructure.
- Staining is a technique used to enhance contrast in biological samples by applying dyes that color specific cells or organelles.
- Detailed cell structure is important to help learners to have a deeper understanding of the structure of cells and their functions.
- Water is essential for all living systems and has unique properties that enable biochemical reactions, stabilise environments for cells, transport substances and support the structure and energetics of life at the cellular level.

TERM 2

CONCEPT 2.3.0: MAINTENANCE OF THE ORGANISM



Introduction

Overview: This concept describes the set of biological activities and systems that allow an organism to live and function properly. These include processes such as feeding, breathing, blood circulation, waste removal, and reproduction, among others. Collectively, they are often referred to as maintenance processes.

Under this concept, the following Sub-topics will be covered: Nutrition- types of nutrition, Autotrophic nutrition, Heterotrophic nutrition and transport in plants.



General competences:

- Analytical thinking
- Collaboration
- Communication
- Critical thinking
- Evaluation
- Problem solving

Hook! 'Survival isn't about growth alone – it is about balance. How do organisms keep equilibrium in a constantly changing environment?'



Key concepts/words/vocabulary

Adenosine triphosphate (ATP) - the universal energy currency of cells

Autotrophs - organisms that produce their own food

Heterotrophs - organisms that consume other organisms to obtain energy

Producers - organisms (like plants, algae, some bacteria) that synthesise their own food

Consumers - organisms (like animals, fungi, some bacteria) that rely on autotrophs or other heterotrophs for food

Sub-Topic 2.3.1 Nutrition- Types of Nutrition.



Introduction:

Overview: Nutrition is an important biological process that enables living organisms to consume food to obtain food nutrients for energy, growth, and maintenance of life. Since different organisms have various ways of acquiring and processing food, nutrition can be classified into different types. This Sub-topic will allow learners to describe nutrition, identify types of heterotrophic nutrition in organisms in the local environment, distinguishing autotrophic nutrition from heterotrophic nutrition as well as relating nutrition to different organisms found in the environment.

Specific Competence: Categorise types of nutrition in living organisms

3.1.1 Describing nutrition (autotrophic and heterotrophic)

Learning materials: Textbooks/handouts/internet source, Notebook and pen, manila paper for group work



Activity 1: Exploring types of nutrition

Procedure:

- i) Put learners into small groups of 4-5
- ii) Ask groups to research on autotrophic nutrition and heterotrophic nutrition
- iii) Ask learners to fill in the information in the table below
- iv) Ask learners to present their findings

Aspect	Autotrophic nutrition	Heterotrophic nutrition
Brief description		
Similarities		
Examples of organisms with this mode of nutrition		

- v) Consolidate learners' findings.

**Content Tip:**

- Both autotrophic and heterotrophic nutrition;
 - require energy in the form of ATP for cellular processes,
 - involve organic molecules (carbohydrates, proteins, fats),
 - rely on respiration to release energy,
 - use enzymes and biochemical reactions to process nutrients and
 - are essential for survival and reproduction

3.1.2 Identifying types of heterotrophic nutrition in organisms in the local environment (parasitism, saprophytic and holozoic)

Learning materials: Notebook and pen, samples of local plants, fungi, and animals, manila paper

**Activity 1: Spotting heterotrophic nutrition in the environment.**

Procedure:

- i) Put learners into small groups of 4-5
- ii) Take learners outside (school yard or nearby area).
- iii) Ask groups to observe and note down examples of organisms they see that might fit into one of the three categories of heterotrophic nutrition; parasitism, saprophytic and holozoic.
- iv) Back in class, ask groups to share their observations
- v) Ask groups to classify each organism into parasitic, saprophytic, or holozoic nutrition
- vi) Consolidate the learners' findings after all groups have presented



Discussion:

- Briefly explain the three types of heterotrophic nutrition

Which type of heterotrophic nutrition is most common in your local environment and why?



Content Tip:

- As learners perform the above activity they should focus on fungi on dead logs, insects feeding on plants, or animals eating food.
- The three types of heterotrophic nutrition are;
 - **Parasitism:** Organisms that live on or inside another organism and harm it (e.g., tapeworms, ticks, lice),
 - **Saprophytic:** Organisms that feed on dead and decaying matter (e.g., mushrooms, molds) and
 - **Holozoic:** Organisms that ingest and digest solid food (e.g., humans, dogs, birds).

3.1.3 Distinguishing autotrophic nutrition from heterotrophic nutrition

Learning materials: cards with names/pictures of organisms (e.g., sunflower, mushroom, lion, human, algae), manila paper, markers



Activity 1: Autotrophs vs. Heterotrophs

Procedure:

- i) Put learners into small groups of 4-5
- ii) Ask learners to collect cards with names/pictures of organisms (e.g., sunflower, mushroom, lion, human, algae).
- iii) Ask learners to sort them into autotrophs and heterotrophs.
- iv) Ask learners to give reason(s) for categorising cards/pictures that way.
- v) Ask groups to present their findings
- vi) Consolidate learners' findings



Discussion:

- Write a short paragraph to compare autotrophic and heterotrophic nutrition.

3.1.4 Relating nutrition to different organisms found in the environment

Learning materials: manila paper, markers, organisms



Activity 1: Observation Walk to relate nutrition to different organisms (plants, insects, birds, fungi, cats, dogs, etc.)

Procedure:

- i) Put learners into small groups of 4-5
- ii) Take learners outside the classroom (school yard, garden, or nearby area).
- iii) Ask learners to list organisms which they are able to see e.g. plants, grass, insects, birds, fungi, etc.
- iv) Back in class, ask learners to classify their observations using the table:

Autotrophs	Heterotrophs

- v) Ask learners to present their findings.
- vi) Consolidate the learners' findings.



Discussion:

- What would happen if autotrophs disappeared from the environment?



Skills Developed:

- Collaboration through group work.
- Communicating the findings through presentations of given tasks.
- Critical thinking through distinguishing autotrophic nutrition from heterotrophic nutrition

Expected Standard: Types of nutrition in living organisms categorised correctly



Assessment

1. A tapeworm is found inside the intestine of a human host. What type of nutrition does it exhibit, and how does this affect the health of the host.
2. A mushroom grows on a decaying log in a school garden. How does this mode of nutrition differ from that of the trees around it, and why is this important for the ecosystem?
3. A rabbit eats grass in the school ground. Which type of nutrition is this and how does it differ with the nutrition of the grass itself?



Sub-Topic 2.3.1 Summary

- Nutrition is the process by which organisms obtain and use nutrients. The two types of nutrition are autotrophic and heterotrophic. Autotrophic nutrition involves organisms manufacturing their own food while heterotrophic nutrition involves organisms obtaining food from autotrophs and other heterotrophs.
- Heterotrophic occurs in different forms such as parasitism, holozoic nutrition and saprophytic nutrition.

Sub-Topic 2.3.2 Autotrophic Nutrition



Introduction

Overview: Autotrophic nutrition is the process by which organisms, such as plants and certain bacteria, produce their own food using simple inorganic substances like carbon dioxide, water, and mineral salts.

In this sub-topic, learners are expected to examine external and internal structures of a leaf, explain the process of photosynthesis in plants, trace the fate of glucose produced by photosynthesis, identify storage organs of the products of photosynthesis investigate factors necessary for the process of photosynthesis in plants, design experiments on factors necessary for photosynthesis in plants, relate how the factors that affect photosynthesis improve crop yield, construct word and chemical equations for photosynthesis process and explain the importance of photosynthesis to the environment.

Specific Competence 1: Analyse phototrophic nutrition in plants

3.2.1 Examining external and internal structures of a leaf

Learning materials: Manila papers, markers, microscope, leaves from a



Activity 1: Examining the external structure of a leaf

Procedure:

- i) Put learners into small groups of 4-5
- ii) Ask learners to collect fresh leaves (e.g., from a guava, or bean plant).
- iii) Ask learners to observe, draw and label the external structures of a leaf. (leaf blade/lamina, midrib, veins, petiole)
- iv) Ask learners to present their findings
- v) Consolidate the learners' findings after all groups have presented.



Discussion:

- Discuss the functions of the following external leaf parts: leaf blade (lamina), midrib, veins, petiole)



Activity 2: Examining the internal leaf structure using a microscope

Procedure:

- i) Put learners into small groups of 4-5
- ii) Ask learners to prepare thin leaf sections
- iii) Ask learners to observe prepared specimen under a microscope starting with low power and then high power magnification
- iv) Ask learners to draw the leaf showing the following internal structures: (upper epidermis, palisade mesophyll, spongy mesophyll, vascular bundles, lower epidermis with stomata)
- v) Ask learners to state the function(s) of the labelled parts
- vi) Ask learners to present their work
- vii) Consolidate learners' findings after all groups have presented



Discussion:

- How do the external and internal structures of a leaf work together to support photosynthesis?

3.2.2 Explaining the process of photosynthesis in plants (light and dark reactions)

Learning materials: Manila paper, markers, hand-outs/internet source, bostik.



Activity 1: Exploring light and dark reaction of photosynthesis.

Procedure:

- Put learners into small groups of 4-5
- Ask the groups to research on the process of photosynthesis, focusing on the light and dark reactions
- Ask the learners to fill in the information in the table below

Process	Description	Where it takes place
Photosynthesis		
Light reaction		
Dark reaction		

- Ask the groups to present their findings
- Consolidate learners' findings after all the groups have presented



Discussion:

- Explain how the light reaction complements the dark reaction in photosynthesis

3.2.3: Tracing the fate of glucose produced by photosynthesis (glucose, sucrose, starch, proteins, lipids)

Learning materials: Labelled cards, manila paper, glue, toothpicks



Activity 1. The journey of glucose in plants

Procedure:

- i) Put learners into small groups of 4-5
- ii) Ask learners to collect cards labeled: Glucose, Sucrose, Starch, Amino acids, Fatty acids, Glycerol, Proteins, and Lipids.
- iii) Ask groups to arrange them into flow diagrams showing how glucose changes form.
- iv) Ask groups presents their diagram and explain the pathway.
- v) Consolidate learners' findings.



Discussion:

- Write a brief paragraph to explain what happens to glucose manufactured by photosynthesis.



Content Tip:

- Glucose is the form in which plants manufacture food, it is transported in form of sucrose then stored as starch.
- Some glucose is converted into fatty acids and glycerol which later combine to form lipids
- Incorporating nitrogen to products formed from glucose forms amino acids.
- Proteins are formed by addition of amino acids.

3.2.4: Identifying storage organs of the products of photosynthesis (leaves, root and stem tubers, fruits, seeds)

Learning materials: Samples of plant storage organs (potato tuber, cassava tuber, maize seeds, beans seeds, mango fruits, spinach leaves ...)



Activity 1: Storage organs in plants

Procedure:

- i) Put learners into small groups of 4-5
- ii) Ask groups to identify and explain how leaves, root and stem tubers, fruits, and seeds act as storage organs for products of photosynthesis.
- iii) Ask groups to present their findings
- iv) Consolidate the learners' findings after all groups have presented



Activity 2: Sorting out challenge- storage organs

Procedure:

- i) Put learners into small groups of 4-5
- ii) Provide each group with pictures or real samples of plant storage organs (potato, cassava, maize, beans, mango, and spinach).
- iii) Ask groups to sort pictures/real samples of plant storage organs into categories: Leaves, Root Tubers, Stem Tubers, Fruits, and Seeds.
- iv) Ask each group to explain why the organ is a storage organ.
- v) Ask groups to make presentations of their findings
- vi) Consolidate learners' findings after all groups have presented



Discussion:

- Write a short paragraph about each plant storage organ *and explain what type of food it stores and why it is important for the plant.*

3.2.5: Investigating factors necessary for the process of photosynthesis in plants.

Learning materials: Textbook/Hand-outs, access to internet, manila paper, markers/pens, note book



Activity 2: Exploring factors necessary for photosynthesis

Procedure

- i) Put learners into small groups of 4-5.
- ii) Ask groups to research on factors necessary for photosynthesis: light energy, carbon dioxide and chlorophyll.
- iii) Ask learners to make presentations to the class
- iv) Consolidate the learners' findings and correct misconceptions



Discussion:

- Why do plants placed in a dark cupboard eventually wither, even if they are watered?

3.2.6: Designing experiments on factors necessary for the process of photosynthesis in plants (Carbon dioxide, light energy, chlorophyll).

Learning materials: Beaker, tripod, gauze, Bunsen burner (or water bath), test tube, ethanol, forceps, white tile, iodine solution, potted plant, black paper or aluminium foil, paper clips, potted plant with variegated leaves (with white and green patches), two potted plants, two bell jars (or large clear plastic bags), potassium hydroxide (KOH) or soda lime, tape.



Activity 1: Testing a leaf for starch

Procedure

- i) Put learners into small groups of 4-5
- ii) Make available all the necessary requirements to the learners.
- iii) Ask learners pluck one leaf from the potted plant and place in boiling water for 30–60 seconds.

- iv) Ask learners to place the leaf in a test tube of ethanol and put the tube in a hot water bath for two minutes until the leaf turns white.
- v) Ask learners to dip the now-brittle leaf back into warm water to soften it.
- vi) Ask learners to spread the leaf on a white tile and add a few drops of iodine solution.
- vii) Ask the groups to make presentations of their findings with the class
- viii) Collaboratively consolidate the learners' findings.

Caution: Turn off any open flames before using ethanol, as it is highly flammable.



Discussion

- Why is the leaf first boiled in water before testing for starch?
- What is the purpose of placing the leaf in alcohol (ethanol) during the experiment?
- What role does iodine solution play in this test?
- What does a blue-black colouration indicate about the presence of starch in the leaf?
- How does this experiment provide evidence that photosynthesis has taken place?
- Why is it important to use a leaf that has been exposed to sunlight for this test?



Activity 2: Light energy as a necessity for photosynthesis

Procedure:

- i) Put learners into small groups of 4-5
- ii) Ask learners to place the potted plant in a dark cupboard/room for 24–48 hours.
- iii) Ask learners to use aluminium foil to cover a portion of one leaf on both sides and secure it with clips.
- iv) Ask learners to place the plant in bright sunlight for 4–6 hours.
- v) Ask learners to pluck the leaf that was partially covered with aluminium foil from the plant and perform the test for starch.
- vi) Ask groups to present their findings to the class

- vii) Consolidate the learners' findings



Discussion:

- Explain why the potted plant is placed in a dark room or cupboard for 24-48 hours before the experiment is carried out.



Content Tip

- Only the uncovered part of the leaf turns blue-black. The covered section remains yellow-brown because it could not receive light to perform photosynthesis, proving light is essential.



Activity 3: Chlorophyll as a necessity for photosynthesis

Procedure:

- Put learners into small groups of 4-5
- Ask learners to place the potted plant in a dark cupboard/room for 24–48 hours.
- Ask learners to place the plant in bright sunlight for 4–6 hours.
- Ask learners to pluck one leaf from the plant and draw the leaf to remember where the green and white areas are.
- Ask learners to perform the test for starch on the leaf.
- Ask groups to present their findings to the class
- Consolidate the learners' findings



Discussion:

- If chlorophyll is essential for photosynthesis, how do plants with differently coloured leaves (such as red or purple) manage to survive and produce food?



Content Tip

- Only the parts that were originally green turn blue-black.
- The white areas lack chlorophyll and therefore cannot produce starch, proving chlorophyll is essential.



Activity 4: Carbon Dioxide as a necessity for photosynthesis

Procedure:

- Put learners into small groups of 4-5
- Ask learners to place two potted plants in a dark cupboard/room for 24–48 hours.
- Ask learners to place each plant under a separate bell jar. Put a beaker of KOH in Jar A and a beaker of water in Jar B.
- Ask learners to seal the jars with a tape and petroleum jelly and place them in sunlight for 4–6 hours.
- Ask learners to open the jars then take a leaf from each plant.
- Ask learners to perform the test for starch on both leaves.
- Ask learners to record their observations in the table below

Setup	Condition	Colour with iodine solution	Conclusion
Plant A			
Plant B			

- Ask groups to present their findings to the class
- Consolidate the learners' findings.



Discussion:

- Why was soda lime (or potassium hydroxide) placed in the container with the plant during the experiment?
- What role does the control plant (exposed to normal air) play in this investigation?
- What would happen to global food chains if atmospheric carbon dioxide levels dropped drastically?



Content Tip

- When setting up the experiment, emphasise that removing or limiting carbon dioxide prevents starch formation in the leaf. For example, placing a plant in a sealed container with soda lime (which absorbs CO₂) will result in leaves that fail to turn blue-black with iodine, unlike control leaves exposed to normal air.
- This clearly demonstrates that carbon dioxide is an essential raw material for photosynthesis.

3.2.7: Relating how the factors that affect photosynthesis improve crop yield.

Learning Materials: Manila paper, markers, access to internet/textbooks, sample of crop images (beans, maize, sunflower, groundnuts....), notebooks and pens.



Activity 1: Crop yield and factors that affect photosynthesis.

Procedure

- i) Put learners into small groups of 4 - 5
- ii) Ask learners to observe two displayed images: one of a healthy maize plant and one of a stunted maize plant.
- iii) Ask learners what differences they notice and give reasons for the observation?
- iv) Ask each group to research the factors affecting photosynthesis in relation to crop yield: Light intensity, carbon dioxide (CO₂) concentration, temperature, water availability and chlorophyll content
- v) Ask groups to make presentations of their findings.
- vi) Collaboratively consolidate the learners' findings



Discussion:

- Analyse which photosynthesis factor might be limiting yield and suggest practical solutions.



Content Tip:

- Light Intensity: High light intensity speeds up photosynthesis, producing more glucose for stronger growth and higher yield.
- Carbon Dioxide: Higher CO₂ (to an optimum) concentration allows plants to make more sugars, increasing biomass and productivity.
- Temperature: Optimal temperatures boost enzyme activity, enhancing photosynthesis and growth; extreme high temperature denatures enzymes hence reducing yield.
- Water Availability: Water supports photosynthesis and keeps stomata open for gaseous exchange, ensuring healthy crop development.
- Chlorophyll Content: More chlorophyll captures more light, improving photosynthesis efficiency and crop yield.

3.2.8: Constructing word and chemical equations for photosynthesis process.

Learning Materials: mixed cards with partially completed equations, manila paper, glue, markers of assorted colours (to highlight reactants vs. products), notebooks and pens.



Activity: Construction of photosynthesis equation.

Procedure:

- i) Put learners into small groups of 4 -5
- ii) Provide learners with incomplete word equation: Carbon dioxide + Water → ? + ? (in the presence of light and chlorophyll)
- iii) Ask learners to complete the word equation.
- iv) Ask learners to construct a balanced chemical equation for photosynthesis.
- v) Give the learners mixed cards: some with reactants, some with products, and some with coefficients.
- vi) Ask groups to arrange cards to form the correct balanced equation on a manila paper.
- vii) Ask each group to check each other's equations.
- viii) Consolidate the learners' work



Content Tip

- The word equation for photosynthesis is carbon dioxide + water → glucose + oxygen (in the presence of light energy).
- Chemical equation for photosynthesis: $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$



Skills Developed:

- Analysing factors that are necessary for photosynthesis.
- Collaboration through group work.
- Communicating the findings through presentations of given tasks.
- Critical thinking by designing experiments on factors affecting the process of photosynthesis in plants.
- Problem solving through relating factors that affect photosynthesis to crop yield.

Expected Standard: Phototrophic nutrition in plants analysed correctly

Specific Competence 2: Evaluate the importance of photosynthesis to the environment

3.2.1: Explaining the importance of photosynthesis to the environment (economic, balance of carbon dioxide and oxygen in the atmosphere, carbon sink....)

Learning materials: Manila paper or projector for group presentations, Hand-outs/Internet access, Markers, notebooks and pens



Activity 1: The role of photosynthesis in the environment

Procedure:

- i) Put learners into small groups of 4-5
- ii) Ask each group to research on importance of photosynthesis in relation to economic importance, balance of carbon dioxide and oxygen in the atmosphere and carbon sinks.
- iii) Ask each group to prepare a short poster or chart on manila paper summarising their findings
- iv) Ask groups to present their findings.
- v) Consolidate the learners' findings and correct their misconceptions.



Discussion

- If global deforestation continues at its current rate, how might the reduction in photosynthesis affect both the economy and the balance of gases in the atmosphere?
- Photosynthesis is often described as a “carbon sink.” In what ways does this role influence climate change, and how could human activities either strengthen or weaken this function?



Content Tip:

- Photosynthesis supports agriculture by producing glucose, the energy source for food chains.
- Trees grow through photosynthesis, supplying wood for industries and livelihoods.
- Energy stored in crops like maize and sugarcane can be converted into renewable fuels.
- Plants absorb carbon dioxide, helping regulate greenhouse gases.
- Oxygen produced during photosynthesis sustains respiration in animals and humans.
- Continuous exchange of gases maintains a stable atmosphere essential for life.



Skills Developed:

- Collaboration through group work.
- Communicating the findings through presentations of given tasks.
- Evaluating the importance of photosynthesis

Expected Standard: Importance of photosynthesis evaluated appropriately



Sub-Topic 2.3.2 Summary

- Autotrophic nutrition is the process by which green plants and some bacteria make their own food.
- In plants, autotrophic nutrition occurs by the process of photosynthesis.
- Plants have leaves that are adapted to carrying out the process of photosynthesis.
- Photosynthesis occurs in two stages; light and dark reaction stages.
- Photosynthesis requires carbon dioxide, light energy and chlorophyll to occur.
- Manufactured food during photosynthesis is stored in organs such as leaves, fruits, tubers and seeds.
- Photosynthesis plays a role in the balance of carbon dioxide and oxygen, provision of food for economic gain.



Assessment



Assignment:

1. Explain the environmental consequences if photosynthesis were to significantly decline worldwide. Assess the impact on atmospheric gases, biodiversity and climate stability.
2. Discuss how photosynthesis contribute to maintaining ecological balance. In your answer, evaluate its role in oxygen production, carbon dioxide regulation and the support of food chains.

Sub-Topic 2. 3.3: Heterotrophic Nutrition



Introduction

Overview: Heterotrophic nutrition is the mode of nutrition in which organisms depend on other organisms for food, as they cannot manufacture their own food. This Sub-topic will allow learners to identify different types of heterotrophic nutrition in living organisms, illustrate types of heterotrophic nutrition in living organisms, as well as evaluate the need for heterotrophic nutrition to organisms and the environment.

Specific Competence 1: Analyse types of heterotrophic nutrition in living organisms

3.3.1: Identifying the different types of heterotrophic nutrition in living organisms (parasitic, saprophytic, holozoic)

Learning materials: Handouts on types of Heterotrophic nutrition/Internet access, markers, manila paper, notebooks and pens.



Activity 1: Types of heterotrophic nutrition

Procedure

- i) Put learners into small groups of 4-5
- ii) Ask each group research on different types of heterotrophic nutrition; parasitic, saprophytic and holozoic
- iii) Instruct learners to focus on the following during their research:
 - **Parasitic Nutrition**
 - Identify types of parasites in plants and animals and give examples of each.
 - Discuss how the parasites harm the host while they are benefiting from them.

- **Saprophytic Nutrition**

- Identify organisms that feed on dead/decaying matter.
- Discuss their role in recycling nutrients.

- **Holozoic Nutrition**

- Identify animals that ingest, digest, absorb, assimilate and egest food.

iv) Ask the groups to make presentations using charts

v) Consolidate the learners' findings and correct misconceptions



Content Tip

- **Parasitic nutrition:** Organisms (parasites) live on or inside a host and derive nutrients at the host's expense. Examples: ticks, tapeworms, lice, and some plants like *Cuscuta*.
- **Saprophytic nutrition:** Organisms feed on dead and decaying organic matter. Examples: fungi (Mushroom), some bacteria. Important for nutrient recycling in ecosystems.
- **Holozoic nutrition:** Organisms ingest, digest, absorb, and assimilate food before egesting waste. Examples: humans and most animals. Involves processes like ingestion, digestion, absorption, assimilation, and egestion.



Discussion:

- Saprophytic organisms like fungi and parasitic organisms such as tapeworms both rely on other sources for nutrition. How do their methods of obtaining food reflect adaptations to their ecological niches, and what advantages or disadvantages might each type of nutrition present in terms of survival and reproduction?

3.3.2: Illustrating Types of Heterotrophic Nutrition in living organisms.

Learning materials: Clear pictures of: A tick feeding on a mammal, Bread mould growing on a slice of bread, Human digestive system, manila paper or digital slides for group presentations, markers.



Activity 1: Illustrating types of heterotrophic nutrition

Procedure

- i) Put learners into small groups of 4 – 5
- ii) Show learners images of a tick on a mammal, bread mould on a slice of bread, and a human digestive diagram.
- iii) Ask learners to predict how each organism obtains food.
- iv) Assign groups to discuss: Parasitic nutrition in ticks, saprophytic nutrition in bread mould, holozoic nutrition in humans. The discussion should be guided by the following questions:
 - How does the organism obtain food?
 - What structural or behavioural adaptations support this mode of nutrition?
 - What are the effects on the host or environment?
- v) Ask learners to create a venn diagram comparing the three types of heterotrophic nutrition.
- vi) Ask groups to present findings.
- vii) Consolidate the learners' findings and correct misconceptions.



Discussion:

Which type of heterotrophic nutrition has the greatest impact on humans, and why?



Content Tip

- Parasitic: Tick sucking blood from an animal
- Saprophytic: Bread mould decomposing organic matter and digesting food externally.
- Holozoic: Human beings digesting food internally



Skills Developed:

- Collaboration through group work.
- Communicating the findings through presentations of given tasks.
- Critical thinking through predicting how different heterotrophs obtain their nutrients.
- Analysing different types of heterotrophic nutrition.

Expected Standard: Types of Heterotrophic nutrition in living things analysed correctly



Assessment

1. Consider a farmer dealing with crop damage caused by parasitic insects and fungal infections. How might understanding the differences between parasitic and saprophytic nutrition help in designing effective control strategies that minimize harm to the ecosystem while protecting crops?

Specific Competence 2: Evaluate the importance of heterotrophic nutrition to organisms and the environment?

3.3.3: Evaluating the need for heterotrophic nutrition to organisms and the environment (obtaining of nutrients; preventing accumulation of dead organic matter in the environment: saprophytes bacteria: sewage treatment, biological control of population, nutrient recycling- Rhizobium, dairy products; fungi: source of food, brewing, manufacture of antibiotics, causing diseases).



Activity 1: The need for heterotrophic nutrition to organisms and the environment.

Learning Materials: Manila paper, markers, hand-outs/access to internet, Sticky notes for peer feedback.

Procedure

- i) Put learners into small groups of 4 – 5.
- ii) Ask each group to research on the following heterotrophs: Saprophytic bacteria, Fungi
- iii) Provide learners with resources to use (Hand-outs/textbooks, internet)
- iv) Each group prepares a poster showing:
 - How the organisms obtain nutrients.
 - The ecological/environmental role of the organisms.
 - Importance of organisms to humans.
- v) Ask groups to display their posters.
- vi) Ask learners to take a gallery walk, to compare each other's posters.
- vii) Consolidate by linking all groups' findings:



Discussion

- Imagine a world where no organism feeds on others or decomposes waste. What would happen to dead plants, animals, and organic matter?
 - Explain how heterotrophic nutrition prevents accumulation of dead matter.
 - Discuss the role of organisms in recycling nutrients back into ecosystems.
 - Describe the economic importance of heterotrophs (food, medicine, industry).



Content Tip:

- Saprophytic bacteria-Role in sewage treatment, biological control, nutrient recycling.
- Fungi - Source of food (edible mushrooms), brewing, production of antibiotics, diseases causing.
- Bacteria – used as a fermenter in dairy-related products such as yogurt and cheese.



Skills Developed:

- Collaboration through group work.
- Communicating the findings through presentations of given tasks.
- Evaluating the importance of heterotrophs.

Expected Standard: Importance of heterotrophic nutrition to organisms and the environment evaluated correctly.



Assessment:

A city's sewage treatment plant suddenly stops functioning because the bacteria involved in breaking down waste are eliminated. Within days, waste begins to accumulate, bad odour spread, and crops nearby start showing poor growth due to lack of recycled nutrients.

- a) Explain how heterotrophic bacteria contribute to sewage treatment and nutrient recycling.
- b) Why is their role essential for both organisms and the environment?



Sub-Topic 2.3.3 Summary:

- Heterotrophic nutrition is the mode of nutrition in which organisms depend on other organisms for food, as they cannot manufacture their own.
- There are three types of heterotrophic nutrition: Parasitic nutrition, Saprophytic nutrition and Holozoic nutrition.
- Heterotrophic nutrition is essential for both organisms and the environment because it.

Sub-Topic 2.3.4: Transport in Plants



Introduction:

Overview: Unlike animals with a heart to pump blood, plants have evolved a special internal transportation system to move water, mineral salts and food nutrients. In this sub-topic, learners will be describing transport system of flowering plants, identifying the organs involved in the transport system in flowering plants, analysing root systems in monocotyledonous and dicotyledonous plants, examining the internal structures of dicot and monocot plants, examining the longitudinal section of xylem and phloem and tracing the uptake and movement of mineral salts and water in the xylem; translocation of nutrients in the phloem

Specific Competence: Demonstrate understanding of transport system of flowering plants

3.4.1: Describing transport system of flowering plants

Learning materials: Food colour, water, beaker, seedlings of beans, hand lenses, scalpel,



Activity 1: The hidden highway

Procedure:

- i) Put learners into small groups of 4-5.
- ii) Ask learners to half-fill two beakers with water.
- iii) Ask learners to add drops of food colour to the water in one beaker (green not preferred)
- iv) Ask learners to uproot two fresh seedlings and wash off the soil from the roots.
- v) Ask learners to place the seedlings in the two beakers respectively.
- vi) Ask learners to place the beakers in direct sunlight for 2-3 hours while observing every 30 minutes.
- vii) Ask learners to record their observations and make presentations.
- viii) Consolidate the findings after the presentations.



Discussion:

- Discuss the difference in the leaf colouration at the end of the experiment.

3.4.2: Identifying the organs involved in the transport system in flowering plants (*root, stem, leaf*)

Learning materials: potted flowering plant, transparent bag, water, food colouring, paper towels, magnifying glass, leaves (bean, lantana, black jack...), sticky notes, manila paper, chart / pictures of vascular bundle and markers.



Activity 1: Identifying organs involved in the transport system in flowering plants - Roots

Procedure:

- Put learners into small groups of 4-5.
- Ask learners to prepare a plant seedling by removing it from the soil to expose the root system.
- Ask each group to make observations on the seedling with exposed roots.
- Ask learners to make presentations.
- Consolidate the learners' findings



Content Tip:

- Visible features to be observed include root hairs, root tip.



Discussion:

- Discuss the functions of the observed features in root function.



Activity 2: Identifying organs involved in the transport system in flowering plants - Stem

Procedure:

- i) Put learners into small groups of 4-5.
- ii) Prepare a simple stem cutting from a flowering plant showing vascular bundles or use already made diagrams or use pictures.
(root system should not be removed)
- iii) Ask each group to collect two real samples of the stem cutting or images.
- iv) Ask learners to half-fill two beakers with water.
- v) Ask learners to add food colour to the water in one beaker.
- vi) Ask learners to place the stem sample into the coloured water and the other stem in clear water, and leave to stand in sunlight for about 2 hours
- vii) Ask learners to compare what happens on the surface of the stems.
- viii) Ask learners to make presentations of their observations
- ix) Consolidate group presentations.



Content Tip

- Vascular bundles become stained with colour
- Stems contain vascular bundle as transport system
- Vascular bundle in stem connect with vascular bundle in roots and leaves.



Activity 3: Identifying organs involved in the transport system in flowering plants – Leaves

Procedure:

- i) Put learners into small groups of 4-5
- ii) Provide each group with a leaf
- iii) Ask learners to observe and record visible features

- iv) Ask learners to make presentations of the observations
- v) Consolidate learners' findings.



Content Tip

- Visible features of leaves should include; the lamina, midrib, branching veins



Discussion:

- Illustrate a simple transport diagram showing the path from soil up to the leaf using a sketched plant.
- If soil is dry and water uptake is limited, which organ(s) would be most affected and why?

3.4.3 Analysing root systems in monocotyledonous and dicotyledonous plants

Learning materials: Potted seedlings of monocots and dicots, beaker, manila, markers, bostik.



Activity 1: Exploring root system in monocots and dicots

Procedure:

- i) Put learners into small groups of 4-5
- ii) Prepare two seedlings (e.g. maize and bean seedling)
- iii) Ask the learners to expose the roots of the two seedlings.
- iv) Ask learners to observe and compare the type of the root system of the two seedlings
- v) Ask learners to record their findings and make presentations.
- vi) Consolidate the learners' findings.



Content Tip

- Features to look for in fibrous and tap root system should include; root length, number of first order lateral roots, density of root hairs, presence of tap root, branching patterns.



Discussion:

- Discuss the advantages of the taproot system over the fibrous root system

3.4.4 Examining the internal structures of dicot and monocot plants (*cross sections of roots and stems of herbaceous and woody plants vascular bundles*)

Learning materials: Permanent slide/ real specimen, microscope, manila, markers

NOTE: If slides are unavailable, alternatives like images can be used



Activity 1: Examine the internal cross-section structure of dicot root in herbaceous plant

Procedure

- Ask learners to work in pairs
- Prepare slides for herbaceous dicot roots or use permanent slides.
- Place microscopes at different work stations.
- Ask learners to observe the given specimen/ permanent slides under the microscope
- Ask learners to draw the observed images
- Ask learners to proceed to activity 2.



Activity 2: Examine the internal cross-section structure of monocot root in herbaceous plant

Procedure

- i) Prepare slides for herbaceous dicot roots or use permanent slides.
- ii) Place microscopes at different work stations.
- iii) Ask learners to observe the given specimen/permanent slides under the microscope
- iv) Ask learners to draw the observed images



Discussion:

Compare the images in activity 1 to that of activity 2



Activity 3: Examine the internal cross-section structure of dicot root in woody plant

Procedure

- i) Ask learners to work in pairs
- ii) Prepare slides for herbaceous dicot roots or use permanent slides.
- iii) Place microscopes at different work stations.
- iv) Ask learners to observe the given specimen/ permanent slides under the microscope
- v) Ask learners to draw the observed images
- vi) Ask learners to proceed with activity 4



Activity 4: Examine the internal cross-section structure of monocot root in woody plant

Procedure

- i) Prepare slides for herbaceous dicot roots or use permanent slides.
- ii) Place microscopes at different work stations.
- iii) Ask learners to observe the given specimen/ permanent slides under the microscope
- iv) Ask learners to draw the observed images



Discussion:

- Compare the images in activity 3 to that of activity 4

3.4.5 Examining the longitudinal section of xylem and phloem

Learning materials: microscope slides, microscope, manila paper, makers.



Activity 1: Examining the longitudinal section of xylem

Procedure

- Ask learners to work in pairs
- Prepare slides for herbaceous dicot roots or use permanent slides.
- Place microscopes at different work stations.
- Ask learners to observe the given specimen/ permanent slides under the microscope
- Ask learners to draw the observed images
- Ask learners to check with other pairs to compare their work.



Discussion:

- How does the longitudinal arrangement of vessels support the continuous transport?



Activity 2: Examining the longitudinal section of phloem

Procedure

- Ask learners to work in pairs
- As learners to prepare slides or use permanent slides for herbaceous dicot roots.
- Place microscopes at different work stations.

- iv) Ask learners to observe the given specimen/permanent slides under the microscope
- v) Ask learners to draw the observed images.



Content Tip

- In dicot vascular bundles are arranged in a ring throughout the stem. In contrast, monocot stems have vascular bundles scattered randomly throughout the ground tissue
- Vascular bundles are important for the transportation of water and nutrients through the xylem. Sugars are transported through the phloem, thereby supporting the plant's overall growth and development
- The cohesion-tension theory explains that water is pulled upward forming a column.
- The phloem, responsible for the translocation of sugars and is located just beneath the bark
- Secondary growth in woody plants is facilitated by the vascular cambium and cork cambium.
- The vascular cambium produces secondary xylem (wood) and secondary phloem, while the cork cambium produces the protective bark.
- Herbaceous plants lack prominent cambial activity, which restricts their ability to produce secondary tissues, hence they remain relatively slender.



Assessment



Assignment:

- Create a tangible model illustrating the longitudinal arrangement of xylem and phloem in a stem, and explain transport direction along the length

3.4.6 Tracing the uptake and movement of mineral salts and water in the xylem; translocation of nutrients in the phloem

Learning materials: Straws or glass tubes (long and narrow), beakers, water, scalpel, tape, healthy wood plant



Activity 1: Mineral salt and water uptake and movement

Procedure

- i) Put learners into small groups of 4-5.
- ii) Ask learners to half-fill a beaker or glass cup with water.
- iii) Ask learners to add 2 drops of food colouring for visibility.
- iv) Ask learners to lower the long narrow tube (straw) into a half-filled beaker or cup of water.
- v) Ask learners to observe the movement of water in the straw.
- vi) Ask learners to make presentations of their observations.
- vii) Consolidate the learners' findings.



Discussion:

- Relate the properties of water to the adaptation of the xylem to conduct water



Content Tip

- Water movement in xylem is made possible by:-
 - End-end structure of the xylem
 - Hollow space in the xylem
 - Capillarity of water
 - Cohesion-tension forces of water



Activity 2: The ring model

Procedure:

- i) Put learners into small groups of 4-5.
- ii) Ask learners to identify a woody healthy plant in the school surrounding
- iii) Ask learners to select a branch on the identified tree.
- iv) Ask learners to carefully make two parallel cuts around the branch of about 2cm apart. (ensuring that the cut does not get deep into the wood)
- v) Ask learners to mark the date and the location of the ring.
- vi) Ask learners to observe the branch over a month.
- vii) Ask learners to make predictions of what will happen to the part of the branch above and below the ring.
- viii) Ask learners to make presentations of their observations after a month.
- ix) Consolidate the learners' findings.



Discussion:

- Justify why there is a difference in girth between the part of the branch above the ring and the part below the ring.



Skills Developed:

- Collaboration through group work.
- Communicating the findings through presentations of given tasks.
- Analysing sections of the root, stem and leaf as dicots or monocots, herbaceous or woody

Expected Standard: Understanding of transport system of flowering plants demonstrated appropriately



Assessment

1. A farmer is planting two types of crops: maize and beans. He notices the roots of the maize are shallow and spread out, while the bean roots go deep and have a main root with branches.
 - (a) Explain the differences in the root system
 - (b) Discuss how these root systems help in water uptake.
2. Mutinta a local subsistent farmer is examining two stem cross-sections under a microscope: one from a sunflower and one from a maize plant.
 - (a) Describe the arrangement of vascular bundles in these stems.
 - (b) Compare the arrangement of vascular bundles in these stems.
3. A fruit grower wants to know why after removing the bark in a ring around the trunk of a tree (girdling) caused the tree to die above the ring.
 - (a) Explain why girdling is harmful.
 - (b) Describe the role of phloem in transporting sugars.
4. Form two learners were passing through a nearby forest. They saw a pine tree and a sunflower plant growing alongside each other. Remembering their lesson on transport system in plants, they challenged each other with tasks. Provide solutions to their tasks.
 - (a) Compare the vascular structures of the pine tree and the sunflower plant.
 - (b) Explain how the pine tree can grow much taller and thicker over time than the sunflower plant.

Specific Competence 2: Analyse the process of transpiration in plants

3.4.7: Describing the process of transpiration

Learning materials: A potted plant, hand-outs/books/internet source, a transparent plastic bag, rubber band, manila paper and markers.



Activity 1: Exploring transpiration

Procedure:

- i) Put learners into small groups of 4-5
- ii) Ask learners to discuss the process of transpiration
- iii) Ask the groups to
 - Write a brief description of transpiration
 - State and explain the forces involved in the movement of water from the soil to the leave
- iv) Ask groups to present their findings to the class.
- v) Consolidate learners' findings after all groups have presented.



Activity 2: Observing transpiration

Procedure:

- i) Put learners into small groups of 4-5
- ii) Ask groups to cover a leaf from a potted plant with the plastic bag and seal it using a rubber band.
- iii) Ask groups to place the plant in sunlight for 3-4 hours.
- iv) Ask groups to write down the observations after 3-4 hours.
- v) Consolidate group findings.



Discussion:

- What do the observation in v) above show about what was happening inside the leaf?
- How can you improve this experiment?

3.4.2 Investigating factors that affect the rate of transpiration

Learning materials: Manila paper, markers, internet source/text books, potted rape seedlings



Activity 1: Exploring the factors that affect the rate of transpiration

Procedure:

- i) Put learners into small groups of 4-5
- ii) Ask the groups to research on the following:
 - Identify the factors that affect the rate of transpiration
 - Describe how each of the identified factor affect the rate of transpiration
- iii) Ask groups to present their findings
- iv) Consolidate the learners' findings after all groups have presented



Activity 2: Effect of temperature on the rate of transpiration

Procedure:

- i) Put learners into small groups of 4-5
- ii) Provide groups with two potted rape seedlings that were not watered the previous day.
- iii) Ask groups to place one seedling near a warm place such as a lamp and another in a cooler place for 2 hours
- iv) Ask groups to write their observation
- v) Ask groups to present their findings
- vi) Consolidate learners' findings after all presentations



Discussion:

- Write a brief paragraph to explain the effect of each of the following factors on the rate of transpiration:
 - Temperature

- Light
- Wind
- Humidity



Content Tip:

- Transpiration increases with increase in temperature. High temperature increases kinetic energy of water molecules.
- High wind speed sweeps away water vapour from the surface of the leaf thereby increasing rate of transpiration.
- A humid environment reduces transpiration because the concentration gradient is lowered.
- High light intensity cause the stomata to open hence increasing the rate of transpiration.

3.4.3 Examining adaptive features in plants that reduce excess loss of water.

Learning materials: Samples or images of different plants (succulents, cactus, pine, banana, etc.)



Activity 1: Plant adaptations that prevent water loss

Procedure:

- Put learners into small groups of 4-5
- Ask the groups to examine pictures of desert plants (cactus, aloe) and tropical plants (banana, hibiscus).
- Ask groups to explain why desert plants look so different from tropical plants
- Ask learners to present their findings
- Consolidate learners' findings after all groups have presented



Activity 2: Exploring adaptive features in plants that reduce excess water loss.

Procedure:

- Put learners into small groups of 4-5

ii) Ask learners to examine leaves and stems of different plants shown on images/ collected from the local environment (if available)

iii) Ask learners to record their observations in a table:

Plant	Feature	How it reduces water loss

iv) Ask learners to present their findings to the class

v) Consolidate learners' findings after all groups have presented



Content Tip:

- As learners perform the above activities, they should note features such as:
- Thick, fleshy succulent stems (e.g. cactus)
- Waxy cuticle on leaves (e.g. cassava)
- Reduced or needle-like leaves (e.g. pine)
- Stomata that open at night Crassulacean Acid Metabolism (CAM) (e.g. aloe vera and other succulents, pine apple, cacti)

3.4.4 Evaluating the significance of transpiration



Activity 1: Investigating the importance of transpiration

Procedure:

- Put learners into small groups of 4-5
- Ask learners to investigate the importance of transpiration

- iii) Ask groups to record identified importance of transpiration
- iv) Ask groups to prepare short explanations on each of the identified importance in iii. above



Content Tip:

- As learners perform the above activities, they should explain the following about the importance of transpiration:
 - Nutrient transport – how transpiration helps move mineral salts from roots to leaves.
 - Cooling effect – how evaporation prevents overheating.
 - Water cycle contribution – how transpiration adds moisture to the atmosphere.
 - Maintaining turgidity – how water movement keeps cells firm.



Skills Developed:

- Collaboration through group work.
- Communicating the findings through presentations of given tasks.
- Evaluating the significance of transpiration.
- Analysing adaptive features of plants to prevent water loss.

Expected Standard: Process of transpiration analysed correctly



Assessment:

1. On a hot sunny day, Musonda notices that the leaves of her tomato plant are wilting. She waters the plant and it recovers.
 - (a) Describe how the water moved from the roots to the leaves then into the atmosphere.
 - (b) Give a scientific explanation of wilting.
 - (c) Explain why the plant wilted.
 - (d) Explain how transpiration is involved in this observation.
 - (e) Evaluate both the benefits and potential disadvantages of transpiration for crops in different seasons.

2. A learner subjected two identical plants to two different conditions:

Plant **A** was kept near a sunny window with moving air.

Plant **B** was kept in a shaded, humid corner of the classroom.

- (a) Predict which plant will lose water faster and explain why.
 - (b) Suggest an experiment to measure the rate of transpiration in both plants.
3. A group of four learners visited a botanical garden and observed cacti plants which had thick stems and spines instead of having leaves similar to other plants at the school garden.
- (a) Justify why the cacti plant appears different from other plants.
 - (b) Identify two features of the cactus that help it survive in dry conditions.
 - (c) Explain how each feature reduces water loss through transpiration.



Sub-Topic 2.3.4 Summary:

- Transpiration is the loss of water vapour from plant leaves through stomata.
- Water is absorbed by roots, transported through xylem, evaporates from mesophyll cells, and diffuses out via stomata.
- This continuous movement creates a **transpiration stream** that helps pull water upward.
- Transpiration is affected by:

Light intensity: Increases stomatal opening, raising transpiration.

Temperature: Higher temperatures speed up evaporation.

Humidity: High humidity slows transpiration; low humidity increases it.

Wind/air movement: Wind removes water vapour around leaves, increasing transpiration.

Leaf surface area: Larger leaves lose more water.

- Adaptive Features in plants that reduce excess water loss

Thick cuticle: Waxy layer reduces evaporation.

Reduced leaf size/spines: Minimizes surface area for water loss (e.g., cactus).

Sunken stomata: Trap moist air, lowering transpiration.

Leaf hairs: Reduce air movement near stomata.

Succulent stems: Store water for dry conditions.

- Significance of Transpiration

Benefits:

- Maintains water movement (transpiration stream).
- Supplies minerals from soil.
- Cools the plant through evaporation.
- Maintains turgor pressure for support.

Disadvantages:

- Excessive water loss can cause wilting.
- In dry conditions, plants risk dehydration.



Concept 2.3.0 Summary:

- Nutrition Types: Organisms obtain nutrients through autotrophic (self-made food, e.g., photosynthesis in plants) or heterotrophic (dependence on other organisms for food) modes.
- Autotrophic Nutrition:
 - Carried out by green plants and some bacteria.
 - Occurs via photosynthesis, which has light and dark stages.
 - Requires carbon dioxide, light energy, and chlorophyll.
 - Food produced is stored in leaves, fruits, tubers, and seeds.

- Helps balance oxygen and carbon dioxide and provides food for economic use.
- Heterotrophic Nutrition: Organisms cannot make their own food.
 - Types include parasitic, saprophytic and holozoic nutrition.
 - Plays an essential role in ecosystems and survival of organisms.
- Transpiration in Plants:
 - Loss of water vapour through stomata.
 - Driven by absorption in roots, transport via xylem, evaporation, and diffusion.
 - Influenced by light, temperature, humidity, wind, and leaf size.
- Adaptations to Reduce Water Loss:
 - Thick cuticle, reduced leaf size/spines, sunken stomata, leaf hairs, and succulent stems.
- Significance of Transpiration:
 - Benefits: Maintains water and mineral flow, cools plants, and supports structure via turgor pressure.
 - Disadvantages: Excessive water loss can cause wilting and dehydration in dry conditions.

CONCEPT 2.4.0: CONTINUITY OF LIFE



Introduction

Overview: The continuity of life is sustained through reproduction and inheritance, enabling species to persist across generations while safeguarding essential genetic traits. Meanwhile, variation within populations promotes adaptation, fueling evolution and ensuring biodiversity in the natural world.



General Competences:

- Communication
- Collaboration
- Analysis
- Environmental sustainability
- Problem solving

Hook! Life never stops; it flows from one generation to the next, like an unbroken chain.



Key concepts/words/vocabulary

- **Asexual reproduction:** Production of new offspring which are genetically identical to the parent plants, without using gametes.
- **Vegetative propagation:** Formation of new plants from vegetative parts (roots, stems, leaves).
- **Grafting:** This is an artificial method of asexual reproduction where a stem cutting (scion) from one plant is joined to the rootstock of another plant so they grow together as one.

- **Budding:** This is a technique where a single bud (scion) from one plant is inserted into the stem of another plant (rootstock).
- **Layering:** This is a method where a branch of the parent plant is bent down and covered with soil until it develops roots. Once rooted, the branch can be cut off and grown as an independent plant.
- **Runners:** Horizontal stems that grow along the ground and form new plants at nodes.
- **Suckers:** Shoots arising from underground stems or roots.
- **Corms:** Swollen underground stems storing food.
- **Tubers:** Swollen stems or roots with buds (“eyes”) that grow into new plants.
- **Bulbs:** Underground storage organs made of fleshy leaves.
- **Rhizomes:** Underground stems that produce shoots and roots.
- **Clones:** Offspring produced by asexual reproduction, genetically identical to the parent.
- **Sexual reproduction:** Production of offspring involving fusion of the male and female gametes; results in genetic variation.
- **Flower:** Reproductive organ of flowering plants.
- **Stamen:** Male reproductive part of a flower, consisting of anther and filament.
- **Anther:** Produces pollen grains (male gametes).
- **Carpel/Pistil:** Female reproductive part of a flower, consisting of stigma, style, and ovary.
- **Ovary:** Contains ovules, which develop into seeds after fertilization.
- **Pollination:** Transfer of pollen from anther to stigma.
- **Fertilization:** Fusion of male and female gametes to form a zygote.
- **Seed:** Structure formed after fertilization, containing the embryo.
- **Fruit:** Mature ovary that protects seeds and aids dispersal.

Sub-Topic: 2.4.1 Reproduction in Plants



Introduction

Overview: Reproduction in plants is the biological process by which new plants are formed, ensuring the survival and continuity of species. This Sub-topic will allow learners to explain the different modes of asexual reproduction in plants, analyse the advantages and

disadvantages of vegetative propagation. The topic will also enable learners to apply different modes of asexual reproduction to propagate different plants.

Specific Competence 1: Propagate different plants using asexual reproduction modes.

4.1.1 Explaining the different modes of asexual reproduction in plants.

Learning material: Textbooks/handouts/internet access, notebook or digital device for recording findings, projector (optional), pens, manila paper, markers, or presentation slides.



Activity 1: Different modes of asexual reproduction in plants

Procedure:

- i) Put learners into small groups of 4-5.
- ii) Ask the groups to research on different modes of asexual reproduction (e.g., runners, rhizomes, tubers, bulbs, suckers, corms, grafting, budding, layering, cuttings, leaf plantlets).
- iii) Ask groups to use the following guides during their research; definition of the mode, plant examples, structure involved and agricultural importance.
- iv) Ask the groups to record their findings in the following table;

Mode of Asexual Reproduction	Plant Example	Structure Used	Key Notes

- v) Ask groups to present their findings to the class using: charts/ PowerPoint slides/oral presentations with visual aids.

- vi) Ask each group to explain the mode of asexual reproduction giving examples and highlight its significance in agriculture.
- vii) Consolidate the learners' findings and correct misconceptions.



Discussion

- Farmers often grow bananas, sugarcane, and potatoes without planting seeds. How do these plants reproduce?
- How do natural and artificial modes of asexual differ in reliability and speed?
- Why is asexual reproduction important for farmers and horticulturists?



Content Tip

- Natural propagation: Runners (Stolons), Rhizomes, tubers, bulbs, corms and suckers
- Artificial propagation: Grafting, budding, layering, and cuttings used in agriculture and horticulture.

4.1.2 Analysing the advantages and disadvantages of vegetative propagation (advantages: increase yields, genetic stability, increased rate in maturity; disadvantages: no genetic variation, plants prone to diseases, overcrowding, undesirable characteristics maintained)

Learning material: Case study hand-outs (cassava, potatoes, bananas), textbooks, internet access, manila paper, markers, notebooks and pens.



Activity 1: Advantages and disadvantages of vegetative propagation

Procedure:

- i) Put learners into two groups.
- ii) Provide both groups with case studies (e.g., cassava farming, potato blight, banana plantations).
- iii) Ask Group A to argue for vegetative propagation (advantages).

- iv) Ask Group B to argue against vegetative propagation (disadvantages).
- v) Ask each group to present their arguments
- vi) Compare arguments from both groups.
- vii) Collaboratively consolidate the facts brought out by both groups.



Discussion

- Farmers in Zambia often grow crops like cassava and sugarcane using stem cuttings instead of seeds.
 - Why might they prefer this method?
 - What risks could it bring?



Content Tip

Advantages:

- Rapid multiplication of plants.
- Offspring are genetically identical to the parent (preserves desirable traits).
- Useful for crops that don't produce viable seeds (e.g., banana, sugarcane).
- Ensures uniformity in yield and quality.

Disadvantages

- Lack of genetic variation makes plants vulnerable to diseases and pests.
- Over-reliance on identical clones reduces adaptability to environmental changes.
- Some methods require skill and resources (e.g., grafting).

4.1.3 Applying different modes of asexual reproduction to propagate different plants (grafting, budding, layering: vegetative structures, runners, suckers, stems, corms, stem tubers, root tubers, bulbs, rhizomes)

Learning material: Healthy rootstock plant (e.g., mango, citrus, or guava seedling, rose seedling, hibiscus, bougainvillea-paper flower, jasmine,), Scion (a stem cutting from a desirable variety of the same species), Vegetative structures (Runners – Strawberry, Suckers – Banana, Stems – Sugarcane, Corms – Crocus or Gladiolus, Stem tubers – Potato, Root tubers – Sweet potato, Bulbs – Onion, Rhizomes – Ginger), Sharp, clean knife or blade, Grafting tape or polythene strip, Labels and markers, Notebook/observation table, pen, Garden soil in pots or directly in the ground, Pegs or stones (to hold the stem in place), Watering can



Activity1: Propagating plants through grafting

Procedure:

- i) Put learners into small groups of 4-5
- ii) Make all the necessary requirements available for the groups to use.
- iii) Ask the groups to select a healthy rootstock and scion of similar thickness.
- iv) Ask the groups to make a slanting cut on the rootstock stem.
- v) Ask the groups to prepare the scion with a complementary cut.
- vi) Ask the groups to join the scion and rootstock so the cambium layers align.
- vii) Ask the groups to secure firmly with grafting tape or polythene strip.
- viii) Ask the groups to label their grafted plant and place it in a shaded area.
- ix) Ask the groups to water the grafted plant regularly and monitor changes.
- x) Ask the groups to record observations in the following table;

Day/Week	Observation Focus	Learner Notes
Day 1	Condition of graft union (alignment, firmness)	
Week 1	Signs of wilting or freshness of scion leaves	
Week 2	Callus (swelling) formation at graft union	
Week 3	New shoot/leaf emergence from scion	
Week 4	Strength of union, overall plant health	
Week 6	Growth rate compared to the plant not grafted	



Time Frames for Observation

- Day 1: Record initial condition of graft union.
 - Weeks 1–2: Monitor scion survival and callus formation.
 - Weeks 3–4: Look for new shoots and leaf growth.
 - Week 6: Assess strength of graft and plant vigor.
- xi) Ask groups to compare each other's grafted plants during the time frame.



Discussion

- If you were a farmer, which crop would you propagate by grafting and why?



Content Tip

- Grafting is an artificial method of asexual reproduction where tissues of two plants are joined so they grow as one.
- For a successful grafting ensure the cambium are accurately aligned, the binding is firm, avoid contamination by touch cut surfaces with bare hands, use proper cut technique, keep the graft moist and monitor regularly.
- Commonly used in fruit trees such as mango, citrus, guava, and apple.



Activity 2: Propagating plants through budding

Procedure:

- i) Put learners into small groups of 4-5
- ii) Make all the necessary requirements available for the groups to use.

- iii) Ask the groups to select a healthy rootstock and a bud from the scion plant.
- iv) Ask groups to make a T-shaped cut on the bark of the rootstock.
- v) Ask the groups to carefully insert the bud into the cut, ensuring cambium contact.
- vi) Ask the groups to wrap the union firmly with budding tape or polythene strip, leaving the bud exposed.
- vii) Ask the groups to label the plant and place it in a shaded area.
- viii) Ask the groups water the plant regularly and monitor changes.
- ix) Ask the groups to record observations in the following table;

Day/Week	Observation Focus	Learner Notes
Day 1	Condition of bud insertion and wrapping	
Week 1	Freshness of bud, signs of wilting	
Week 2	Callus formation around bud union	
Week 3	Swelling of bud, signs of growth	
Week 4	Emergence of new shoot from bud	
Week 6	Strength of union, overall plant health	



Time Frames for Observation

- Day 1: Record condition of bud insertion.
 - Weeks 1–2: Monitor bud survival and callus formation.
 - Weeks 3–4: Look for swelling and new shoot emergence.
 - Week 6: Assess strength of union and plant vigor.
- x) Ask groups to compare each other's plants during the time frame.



Discussion

- If you were a farmer, which crop would you propagate by budding and why?

Content Tip

- Budding is an artificial method of asexual reproduction where a single bud (scion) from one plant is inserted into the stem of another plant (rootstock) so they grow together.
- Precautions in budding are similar to grafting.
- A bud from the scion is carefully placed into the cut.
- The union is wrapped with tape, leaving the bud exposed.
- Over time, the bud grows into a new shoot while attached to the rootstock.
- Commonly used in fruit trees (citrus, peach, plum) and ornamental plants like roses.



Activity 3: Propagating plants by layering

Procedure:

- i) Put learners into small groups of 4-5
- ii) Make all the necessary requirements available for the groups to use.
- iii) Ask the groups to select a healthy, low-growing branch of the parent plant.
- iv) Ask the groups to bend the branch down to the soil surface.
- v) Ask the groups to make a small slit or wound on the underside of the branch (optional, to encourage rooting).
- vi) Ask the groups to cover the wounded part with soil, leaving the tip of the branch exposed.
- vii) Ask the groups to secure the branch with a peg or stone.
- viii) Ask the groups to water regularly and label the layered branch.
- ix) Ask the groups to monitor changes over several weeks.
- x) Ask the groups to record observations in the following table;

Day/Week	Observation Focus	Learner Notes
Day 1	Condition of branch placement and soil cover	
Week 1	Freshness of leaves, signs of wilting	
Week 2	Swelling or callus formation at buried section	
Week 3	Initial root development (if visible)	
Week 4	Stronger root growth, new shoots emerging	
Week 6	Ability to cut and transplant the new plant	

Note: When the layered part develops its own roots, it can be cut off from the main plant.



Time Frames for Observation

- Day 1: Record condition of branch placement and soil cover.
- Weeks 1–2: Monitor leaf freshness and callus/root initiation.
- Weeks 3–4: Look for root development and new shoots.
- Week 6 onwards: Assess readiness for separation and transplantation.

xi) Ask groups to compare each other's plants during the time frame.



Discussion

- If you were a farmer, which local crop would you propagate by layering and why?



Content Tip

- Layering is a natural or artificial method of asexual reproduction where a stem of the parent plant is bent down and covered with soil until it develops roots, after which it can be separated to grow independently.

- When conducting layering precautions include choosing a healthy plant with flexible branches, avoid deep cuts, maintain consistent moisture but avoid water logging, target layering during the growing season, keep tools clean to avoid infections.
- Commonly used in plants with flexible stems such as hibiscus, bougainvillea, jasmine, and strawberry.



Activity 4: Propagating plants through vegetative structures

Procedure:

- Put learners into small groups of 4-5
- Make all the necessary requirements available for the groups to use.
- Ask the groups to collect plant samples representing each vegetative structure.
- Ask the groups to plant the vegetative structures in prepared soil under similar conditions.
- Ask the groups to label each sample clearly with its vegetative structure type.
- Ask the groups to water regularly and monitor growth.
- Ask the groups to record observations at set intervals in the following table.

Day/Week	Structure	Observation Focus	Learner Notes
Day 1	All	Condition of planting material	
Week 1	Runners, Suckers	Freshness of shoots, signs of wilting	
Week 2	Tubers, Bulbs, Rhizomes	Swelling, sprouting buds	
Week 3	Stems, Corms	Emergence of new shoots/leaves	
Week 4	All	Growth rate, leaf development	
Week 6	All	Strength of new plants, readiness for transplant	

- Ask the groups to compare their plants after six weeks.



Time Frames for Observation

- Day 1: Record condition of planting material.
- Weeks 1–2: Monitor survival, sprouting, and callus formation.
- Weeks 3–4: Look for new shoots, leaves, and root development.
- Week 6: Assess plant vigor and readiness for transplanting.



Discussion

- Why do some vegetative structures (e.g., stem tubers like potato) sprout faster than others (e.g., suckers in banana)?
- What environmental or structural factors influence the difference in the rate of sprouting in some plants?
- If a farmer wants to quickly multiply crops with uniform quality, which vegetative structure would be most suitable and why?



Content Tip

- Runners: Horizontal stems that root at nodes (e.g., *strawberry*).
- Suckers: Shoots from underground stems or roots (e.g., *banana*).
- Stems: Cuttings used for crops like *sugarcane*.
- Corms: Swollen underground stems (e.g., *crocus*).
- Stem Tubers: Swollen stems with buds (“eyes”) (e.g., *potato*).
- Root Tubers: Swollen roots storing food (e.g., *sweet potato*).
- Bulbs: Short stems with fleshy leaves (e.g., *onion*).
- Rhizomes: Underground stems that sprout shoots (e.g., *ginger*).



Skills Developed

- Collaboration through group work.
- Communicating the findings through presentations of given tasks.
- Analysing modes of vegetative propagation.
- Environmental sustainability through conserving plants as investigations are being done.
- Problem solving through selecting the correct specimen.

Expected Standard: Different plants propagated appropriately



Assessment

1. A school owns a small farm where they grow strawberries. They notice that new strawberry plants keep appearing around the mother plant without them planting new seeds. These new plants are connected to the parent plant by thin stems that creep along the soil.
 - (a) Identify the natural mode of vegetative propagation occurring in Maria's farm.
 - (b) Explain how this method benefits the farmer in terms of crop yield and uniformity.
 - (c) What could be a limitation of relying only on this method for large-scale farming?
2. Monde, a horticulturist, wants to grow rose plants quickly for commercial sale. Instead of waiting for seeds, she cuts a healthy stem from a mature rose plant and places it in moist soil. After a few weeks, roots begin to develop from the cut stem.
 - (a) Name the artificial method of vegetative propagation John used.
 - (b) Why is this method preferred over seed propagation for roses?
 - (c) Suggest another artificial method John could use for propagating roses and briefly describe how it works.
3. A Form 2 agriculture club is experimenting with growing potatoes and sugarcane. They use tubers for potatoes and stem cuttings for sugarcane.

- (a) Classify each crop's propagation method as natural or artificial.
 - (b) Compare the advantages of using tubers versus stem cuttings.
 - (c) If the club wants to ensure genetic diversity in their crops, which method (seed or vegetative propagation) should they consider, and why?
4. A farmer notices that his banana plantation propagated through suckers is being attacked by a fungal disease. Since all plants are genetically identical, the disease spreads rapidly.
- (a) Explain why vegetative propagation made the plantation more vulnerable to disease.
 - (b) Suggest one way the farmer could reduce this risk in future plantations.
 - (c) How does this scenario highlight a disadvantage of vegetative propagation compared to seed propagation?

Specific Competence 2: Analyse sexual reproduction in flowering plants.

4.1.4 Describing sexual reproduction in flowering plants

Learning material: Biology textbooks/internet access, manila paper, markers, notebooks, pens, hand-outs for guiding research (e.g., Stages of sexual reproduction, Pollination agents, Seed formation)



Activity 1: Sexual Reproduction in Flowering Plants

Procedure:

- i) Put learners into small groups of 4 - 5.
- ii) Ask the groups to research on sexual reproduction in flowering plants
- iii) Instruct the groups to use the table below as a research guide

Topic	Key Points	Learner Notes
Flower	• Role of the flower in sexual reproduction	
Fusion of Gametes	• Male and female gametes produced by flowers. • Product of the fusion of gametes	
Significance of Sexual Reproduction in Plants	• Explain genetic variation, survival, adaptation, and continuity of species	

iv) Ask each group to make a presentation of their findings

v) Collaboratively consolidate the learners' findings and correct misconceptions.



Discussion

- How do flowering plants reproduce to ensure survival of their species?
- How do flower structures support their reproductive roles?
- Briefly outline the importance of sexual reproduction in plants.



Content Tip

- Flowers are the reproductive organs of flowering plants.
- During sexual reproduction in plants pollen grains and ovules fuse to form a zygote which develops into a seed.
- Sexual reproduction introduces genetic variation, helping plants adapt to changing environments. It ensures species continuity, survival, and evolution.

1.1.5: Collecting flowers from different plant species found in the local environment

Learning material: Flowers, small scissors, paper bags or envelopes for storing collected flowers, labels and markers, hand lens, manila paper or notebooks and pens.



Activity 1: Collecting and studying flowers from local plant species

Procedure

- i) Put learners into small groups of 4 -5
- ii) Ask learners to collect flowers from at least 3-5 different plant species in the local environment.
- iii) Instruct learners to carefully collect flowers without damaging the plant.
- iv) Ask learners to label each flower with the plant's name (if known) or a description of where it was found.
- v) Ask learners to use hand lenses and examine the structure of each flower.
- vi) Ask learners to record observations in the following table:

Name of flower/plant	Flower Colour	Special Features

- vii) Ask groups to prepare a chart or short presentation comparing the flowers collected.
- viii) Ask learners to make presentations of their findings
- ix) Collaboratively consolidate the learners' findings



Discussion

- Why are flowers important in sexual reproduction of plants?
- Explain how the diversity of flowers in our environment supports plant reproduction and survival.



Content Tip

- Flowers are the reproductive organs of plants.
- Diversity in flower structure reflects adaptations to different pollination methods (insects, wind, birds).

4.1.6 Identifying the male parts (stamen: filament, anther); and female parts (carpel: stigma, style, ovary, ovules) of a flower

Learning material: Fresh flowers (e.g., hibiscus, lily, or any locally available species), hand lens, scalpel or scissors for careful dissection, manila paper or notebooks, observation tables, reference material.



Activity 1: Identifying Male and Female Parts of a Flower

Procedure:

- i) Put learners into small groups 4 - 5
- ii) Make all the necessary requirements available for the groups to use.
- iii) Ask each group to dissect a flower carefully to expose reproductive structures.
- iv) Ask learners to identify male parts (stamen: filament, anther) and female parts (carpel: stigma, style, ovary, ovules).
- v) Ask learners to use a hand lens to observe and record details in the following tables.

Table 1: Male parts (stamen)

Part	Description of part

Table 2: Female parts (carpel)

Part	Description of part

- vi) Ask each group to make presentations of their findings.
- vii) Consolidate the learners' findings



Discussion

- Explain how the male and female parts of a flower work together to ensure sexual reproduction in plants.



Content Tip

- Male Parts (Stamen)
 - Filament: A slender stalk that supports the anther.
 - Anther: Usually oval or lobed, located at the tip of the filament, containing pollen grains.
- Female Parts (Carpel/Pistil)

- Stigma: The top part of the carpel, often sticky or feathery, positioned to receive pollen.
- Style: A narrow tube-like structure connecting the stigma to the ovary.
- Ovary: A swollen basal part of the carpel that encloses ovules.
- Ovules: Small structures inside the ovary, each capable of developing into a seed after fertilization.

4.1.7: Observing, drawing and labelling parts of a typical flower.

Learning material: Fresh flowers (e.g., hibiscus, lily, or any locally available species), hand lens or magnifying glass, Scalpel or scissors for careful dissection, manila paper or notebooks, pencils, rulers, and colored pencils for drawing, observation tables.



Activity 1: Observing, drawing, and labelling parts of a flower.

Procedure:

- Put learners into small groups of 4 - 5.
- Make all the necessary requirements available for the groups to use.
- Ask each group to carefully dissect a flower to expose its structures.
- Ask the groups to use a hand lens and observe the parts of the flower
- Ask the groups to draw the flower on their manila papers/notebooks, showing both external and internal structures.
- Ask the groups to label each part clearly.
- Ask learners to fill in the observation tables with notes and sketches.

Observation Tables

Table 1: External parts of the flower

Part	Description	Sketch
Petals		
Sepals		
Pedicel		

Table 2: Reproductive parts of the flower

Part	Sketch
Filament	
Anther	
Stigma	
Style	
Ovary	
Ovules	

viii) Ask each group to make presentations of their findings.

ix) Consolidate the learners' findings



Discussion

- What differences did you notice between external and internal parts of the flower?
- How do the male and female parts complement each other in reproduction?
- Why is it important to draw and label flowers accurately in biology?



Content Tip

- The flower's external and internal structures are first carefully examined using the naked eye or a hand lens.
- A neat, clear and realistic diagram of the flower is made using simple outlines. Shading should be avoided.
- Label the main parts of the flower accurately. These include:
 - External parts: petals, sepals, pedicel.
 - Male parts (stamen): filament, anther.
 - Female parts (carpel): stigma, style, ovary, ovules.
- Use straight lines for labels, written horizontally, and keep the diagram uncluttered.

4.1.8: Explaining the functions of the parts of a typical flower (monocot and dicot flower).

Learning material: Biology textbooks, internet access, fresh monocot flowers (e.g., lily, maize tassel) and dicot flowers (e.g., hibiscus, sunflower), hand lens, manila paper, markers, observation tables.



Activity 1: Functions of flower parts in monocot and dicot flowers

Procedure:

- i) Put learners into small groups of 4 – 5
- ii) Ask learners to research on the functions of flower parts in both monocots and dicots.
- iii) Ask learners to compare monocot and dicot examples and record findings.
- iv) Ask each group to prepare a chart showing their findings.
- v) Ask the groups to make presentations of their findings; presentations must include diagrams, functions of different plant parts, and examples of monocot and dicot flowers.
- vi) Consolidate the learners' findings and correct misconceptions



Discussion

- Explain how the functions of flower parts in monocot and dicot flowers ensure successful sexual reproduction.
- Why is it important to understand the differences between monocot and dicot flowers?

4.1.9: Describing the process of pollination

Learning materials: Internet source/ books, manila paper, markers



Activity 1: Understanding Pollination

Procedure:

- i) Put learners into small groups of 4-5
- ii) Ask the groups to research on the following:
 - Description of pollination
 - Explanation of the main types of pollination
 - Explanation of the importance of pollination
- iii) Ask learners to copy and fill the information in the table below

	Brief description	Importance
Pollination		

Types of pollination	Brief description	Agent of pollination
1		
2		

- iv) Ask learners to present their findings to the class
- v) Consolidate the learners' findings after all groups have presented

4.1.10: Deducing the type of pollination based on the physical features of the flower (cross pollination and self-pollination)

Learning materials: Pictures or real samples of different flowers (pea plant, hibiscus, maize, sunflower...)



Activity 1: Identifying types of pollination

Procedure:

- i) Put learners into small groups of 4-5
- i) Make all the necessary requirements available for the groups to use.

- ii) Ask groups to examine features and predict whether a flower is self-pollinated or cross-pollinated
- iii) Ask learners to fill in the table below
- iv) Ask learners to present their findings
- v) Consolidate learners' findings after all groups have presented

Feature	Cross-pollination	Self-pollination
Flower size		
Position of anther and stigma		
Petals and nectaries		
Pollination agents		
Genetic variation		
Examples		

4.1.11: Deducing the agents of pollination based on the features of the flower (wind pollinated and insect pollinated)

Learning materials: manila paper, pictures or real samples of flowers (maize, grass, hibiscus, sunflower).



Activity: Relating the flower features to pollination agents

Procedure:

- i) Put learners into small groups of 4-5
- ii) Make available pictures or real samples of flowers to the learners.
- iii) Ask learners to examine features of flowers and predict whether the flower is wind-pollinated or insect-pollinated.
- iv) Ask groups to record their findings and presents to the class.
- v) Consolidate learners' findings after all groups have presented



Discussion:

- Imagine you are a botanist working in a conservation area. You are tasked with explaining to a group of farmers why certain crops (like beans and sunflowers) produce fewer pollen grains compared to grasses such as maize. The farmers are curious because they notice that their bean flowers attract bees, and then maize flowers release large amounts of pollen into the air. Using this scenario, respond to the farmers by:
 - Explaining why insect-pollinated flowers often produce fewer pollen grains compared to wind-pollinated ones.
 - Giving at least two examples of crops that rely on insect pollination and two that rely on wind pollination.
 - Suggesting one advantage and one disadvantage of each pollination method for farmers.
 - Explaining the significance of the bee being attracted by the flower.



Content Tip:

- Insect-pollinated flowers produce fewer pollen grains because insects transfer pollen directly to another flower, making pollination more efficient.
- Wind-pollinated flowers produce large amounts of pollen since wind dispersal is random and less targeted.

4.1.12: Illustrating the path taken by a pollen tube leading to double fertilisation of polar nuclei and egg cell

Learning materials: Books/ internet source, manila paper, print-outs of flowers clearly showing the carpel



Activity 1: Illustrating the path followed by a pollen tube

Procedure:

- i) Put learners into small groups of 4-5

- ii) Ask learners to research on the events leading to double fertilisation
- iii) Ask learners to track the path of the pollen tube, step by step, using the diagram of a flower showing stigma, style, ovary and ovule.
- iv) Consolidate learners' findings after presentations



Content Tip:

- Path of the pollen tube:
 - Pollen grain germinates on stigma after pollination.
 - Tube grows through style.
 - Tube enters ovule through micropyle.
 - Two male gametes travel down tube.
 - One fertilises egg cell, the other fertilises polar nuclei.



Discussion:

- Design a **flowchart** showing the pollen tube's journey and double fertilisation.
- Display your work as a classroom gallery

Specific Competence 3: Deduce the methods of seed and fruit dispersal

4.1.13: Collecting different seeds and fruits in the local environment

Learning materials: Notebook, pen, assorted fruits and seeds.



Activity 1: Fruit and seed collection

Procedure

- i) Put learners into small groups of 4-5.
- ii) Ask learners to identify and collect assorted fruits and seeds in the local environment.
- iii) Ask learners to compare the fruits and seeds in terms of shape, colour, texture, scent, appearance etc.
- iv) Ask learners to make presentations of their findings.
- v) Consolidate learners' findings.

4.1.14: Comparing and contrasting features of collected seeds and fruits in relation to the method of dispersal

Learning materials: Assorted fruits and seeds, feature cards



Activity 1: Fruit and seed sort

Procedure:

- i) Put learners into small groups of 4-5
- ii) Using the collected fruits, ask learners to match the fruits with the feature card accordingly
- iii) Ask learners to complete the table after matching the fruits and seeds with the feature card.

Name of fruit / seed	Feature

- iv) Ask learners to make presentations of their findings.
- v) Consolidate the learners' findings.

4.1.15: Predicting the method of seed and fruit dispersal based on seed and fruit structure

Learning materials: Assorted fruits and seeds, manila, marker, bostik.



Activity 1: Relating fruit and seed characteristics to mode of dispersal

Procedure

- i) Put learners into small groups of 4-5.
- ii) Using their findings from the previous activity, ask learners to predict with explanation the method of dispersal for the seeds and fruits.
- iii) Ask learners to fill in the table as given.

Name of fruit/seed	Characteristic features	Mode of dispersal	Justification

- iv) Ask learners to present their findings.
- v) Consolidate the learners' findings.



Content Tip

- Dispersal is the process by which seeds or fruits are moved away from the parent plant to new locations where they can germinate and grow.
- Adaptive features for dispersal include fleshy fruit, presence of hooks or sticky coatings, shape (e.g. parachute shape), lightweight, hard shell or hard testa, colour, hairy, scent, etc.
- Fruits and seeds are dispersed by wind, water, animals and by self.



Skills Developed

- Communicating findings
- Collaboration through group work
- Analytical thinking through relating the fruits and seeds' adaptive features to the mode of dispersal.
- Critical thinking by making predictions and justification for the modes of dispersal.

Expected Standard: Methods of seed and fruit dispersal deduced accordingly



Assessment

1. A group of form 1 learners collected three different fruits and seeds from their local environment. Out of curiosity, they wanted to know why the seeds and fruits had different features as outlined. Upon presenting the seeds and fruits to you, you had to explain to them that the features were meant for the efficient dispersal:
 - Fruit A: A light, dry seed with thin, papery wings attached.
 - Fruit B: A fleshy, brightly colored fruit with a sweet scent.
 - Fruit C: A hard, buoyant seed with a fibrous husk.

Help these learners to:

- a) Predict the method of dispersal for each fruit/seed.
- b) Explain how the structural features support your prediction.



Sub-Topic 2.4.1 Summary

- Modes of asexual reproduction: Vegetative propagation - new plants grow from vegetative structures such as stems, roots, or leaves. Common methods include; grafting, budding, layering and use of natural structures (runners, suckers, stems, corms, tubers, bulbs and rhizomes).
- Advantages of vegetative propagation include; increased yields, genetic stability (offspring identical to parent) and faster maturity compared to seed-grown plants.
- Disadvantages of vegetative propagation include; no genetic variation (limits adaptability), plants are more prone to diseases and pests, overcrowding can occur and undesirable characteristics are maintained.
- Plant Propagation can be applied in grafting fruit trees, budding roses, layering in shrubs etc.
- Sexual reproduction in flowering plants involves the fusion of male and female gametes. Flowers are the reproductive organs of angiosperms.
- A flower has the male parts (stamen)-filament, anther and the female parts (carpel) - stigma, style, ovary, and ovules. Each of these parts has a role in reproduction.
- Pollination involves transfer of pollen from anther to stigma. There are two types; self and cross pollination.
- Agents of pollination are wind and insects. Flowers are adapted to the type of agent for efficient pollination.
- In flowers double fertilisation occurs where one sperm nucleus fuses with egg cell, another with polar nuclei.
- After fertilisation a fruit with seeds develops.
- Seeds and fruits are dispersed from the parent plant by wind, self, water or animals. Each seed or fruit have adaptive features suitable for the mode of dispersal.

Sub - Topic 2.4.2: Germination, Growth and Development in plants



Introduction:

Overview: Germination is the process by which a seed develops into a new plant. In flowering plants, growth of a plant begins with the seed which contains the embryo. In this sub - topic learners will be describing germination in flowering plants, investigating conditions necessary for germination, distinguishing hypogeal and epigeal germination in plants, describing growth and development in plants, and examining regions of growth using microscopes and micrographs.

Specific Competence: Explore germination, growth and development in plants

4.2.1: Describing germination in flowering plants

Learning materials: Fast germinating seeds (maize, beans ...), cotton wool/paper towels, transparent glass jars/plastic cups, water, labels, markers, notebooks and pens.



Activity 1: Describing germination.

Procedure:

- i) Put learners into small groups of 4-5.
- ii) Ask learners to place cotton wool or folded paper towels inside each jar/cup.
- iii) Ask learners to moisten the cotton wool/paper towel with water (not soaking).
- iv) Ask learners to place 2–3 seeds on the moist cotton/paper. Label each jar with the date and seed type.
- v) Ask learners to keep jars in a warm, well-lit area (but not direct harsh sunlight). They should be adding small amounts of water daily to maintain moisture.
- vi) Ask learners to observe and record results over a period of one week.
- vii) Ask learners to make presentations.
- viii) Consolidate the learners' findings after presentations.



Discussion:

- Relate the observed results with biological processes.



Content Tip:

- Learners to look out for seeds absorbing water (imbibition) observed by increase in size, radicle emerging first (anchoring the seed) then plumule growing upward (forming the shoot).
- Imbibition triggers enzymes.
- The stored food in the cotyledons/endosperm fuels growth.
- The emergence of radicle and plumule marks the start of independent growth.

4.2.2: Investigating conditions necessary for germination (*moisture, suitable temperature and oxygen*)

Learning materials: Fast germinating seeds (maize, beans ...), cotton wool/paper towels, transparent glass jars/plastic cups, water, labels, markers, Access to different temperature environments (e.g., refrigerator, room temperature, warm area near sunlight or incubator), notebooks and pens.



Activity 1: Investigating water as a necessity for germination

Procedure:

- Put learners into small groups of 4-5.
- Make the necessary materials available to the learners. (2 jars, cotton wool, water, 2 sets of seeds)
- Ask learners to set up the jars with labels as follows:
 - Jar A: Place moist cotton wool/paper towel inside and add 3–4 seeds.
 - Jar B: Place dry cotton wool/paper towel inside and add 3–4 seeds.
- Ask learners to place both jars in a warm and well lit environment.
- Ask learners to keep the cotton wool in Jar A moist by watering it every day.
- Ask learners to make observations for a week and record in tabular form as shown.

	Day 2	Day 4	Day 6	Day 7
Jar A				
Jar B				

- vii) Ask learners to make presentations.
- viii) Consolidate the learners' findings after the presentations.



Assignment

1. Ask learners to repeat the procedures by varying amounts of water (too little, optimal, too much) to see effects on germination.



Activity 2: Investigating that a suitable temperature is a necessity for germination

Procedure:

- i) Put learners into small groups of 4-5.
- ii) Ensure all the materials needed are available (seeds, water, thermometer, cotton wool)
- iii) Ask learners to set the experiment as below:-
 - Jar A: Place moist cotton wool/paper towel and seeds. Keep in a cool environment (e.g. refrigerator).
 - Jar B: Place moist cotton wool/paper towel and seeds. Keep at room temperature (moderate).
 - Jar C: Place moist cotton wool/paper towel and seeds. Keep in a warm environment (incubator or insulated box).
- iv) Ask learners to record their observations over a period of one week. Ensure to add water to each jar to keep the cotton wool moist.

	Day 2	Day 4	Day 6	Day 7
Jar A				
Jar B				
Jar C				

- v) Ask learners to make presentations.
- vi) Consolidate learners' findings after the presentations.



Activity 3: Investigating oxygen as a necessity for germination

Procedure:

- i) Put learners into small groups of 4-5.
- ii) Make available all the materials needed.
- iii) Ask learners to make a set up as:
 Jar A: Place moist cotton wool/paper towel inside, add seeds, and leave the jar open to air.
 Jar B: Place moist cotton wool/paper towel inside, add seeds, and seal the jar tightly with a lid or plastic wrap to restrict airflow.
- iv) Ask learners to Place both jars in the same warm environment, adding water daily to maintain moisture.
- v) Ask learners to observe and record results over a period of 7 days.

	Day 2	Day 4	Day 6	Day 7
Jar A				
Jar B				

- vi) Ask learners to make presentations.
- vii) Consolidate learners' findings after the presentation.

4.2.3: Distinguishing hypogeal and epigeal germination in plants

Learning materials: Fast germinating seeds (known for epigeal - beans and hypogeal - maize germination respectively), cotton wool/paper towels, petri dishes, water, labels, markers, notebooks and pens.



Activity 1: Exploring types of germination.

Procedure:

- i) Put learners into small groups of 4-5.
- ii) Ask learners to prepare the petri dishes with a growth medium by lining it with moist cotton wool.
- iii) Ask learners to place the seeds in the moist cotton wool in different petri dishes and label them A and B
- iv) Ask learners to observe and record the results over a period of one week.
- v) Ask learners to compare the results for seeds in A and B
- vi) Ask learners to make presentations.
- vii) Consolidate learners' findings after presentations.



Content Tip:

- Characteristics of epigeal germination: cotyledons lifted above the soil surface, pushing cotyledons upwards, and cotyledons are exposed.
- Characteristics of hypogeal germination: cotyledons remain underground, pushing the plumule above the soil.

4.2.4: Describing growth and development in plants (*primary and secondary growth*)

Learning materials: Fast growing potted plants, water, ruler, notebooks and pens.



Activity 1: Primary and secondary growth in plants.

Procedure:

- i) Put learners into small groups of 4-5.
- ii) Using simple diagrams, illustrate the concepts of primary and secondary growth.
- iii) Make available fast growing potted plants to the learners.
- iv) Ask learners to measure stem length, number of nodes, and stem thickness (optional) at regular intervals for a period of 2-3 weeks.
- v) Ask learners to label which growth is elongation and which is thickening (if visible later).
- vi) At the end of 3 weeks, ask learners to create a mini-poster showing a timeline of growth phases with photos or drawings.

- vii) Ask learners to make presentations.
- viii) Consolidate learners' findings after presentations.



Content Tip:

- After germination, plants grow through two processes:
 1. Primary growth which occurs by lengthening roots and shoots. This allows the plant to explore new soil and reach out to sunlight.
 2. Secondary growth which occurs in woody plants by thickening stems and roots, where the vascular cambium produces new xylem and phloem. This strengthens the plant and improves transport of water and nutrients.
- Under the microscope, distinct zones of growth in roots and shoots can be seen as: Cell division zone, elongation zone and differentiation zone:

4.2.5: Examining regions of growth using microscopes, micrographs (*cell division, elongation, and differentiation into phloem and xylem and cambium*)

Learning materials: Permanent slides/micrographs, microscopes, manila paper, markers and bostik.



Activity 1:

Procedure:

- i) Put learners into small groups of 4-5.
- ii) Ask learners to examine the slides/micrographs to locate regions of active cell division, elongation and differentiation
- iii) Ask learners to observe and draw sketch diagrams showing these regions in a root and shoot respectively.
- iv) Ask learners to make presentations.
- v) Consolidate learners' findings after presentations.



Assignments

1. Distinguish between cell elongation zones and differentiation zones in stems and roots using micrographs.
2. Explain how cells change from generic meristematic cells to specialized tissues.



Skills Developed

- Communicating findings
- Collaboration through group work
- Analytical thinking through comparing types of germination

Expected Standard: Germination, growth and development in plants explored accordingly.



Sub-Topic 2.4.2 summary

- Germination is the process by which a seed develops into a new plant.
- Two types of germination are epigeal and hypogeal.
- Conditions needed for germination to occur are oxygen, water and suitable temperature.
- Growth is in two phases: primary and secondary.
- Zones of growth include cell division, elongation and differentiation.



Concept 2.4.0 summary

- Plants can reproduce without seeds through vegetative propagation.
- Common methods: Cuttings (e.g., cassava, hibiscus), runners (e.g., strawberry), tubers (e.g., potato) and bulbs (e.g., onion)
- Advantages of vegetative propagation: fast, identical offspring, useful in agriculture.
- Disadvantages of vegetative propagation: no genetic variation, making plants vulnerable to diseases.
- Sexual reproduction in flowering plants involves flowers as reproductive organs.
- Key processes of sexual reproduction in flowering plants are: pollination and fertilization
- Sexual reproduction in flowering plants results in seed and fruit formation.
- Seed and fruit dispersal prevents overcrowding and competition by ensuring wide distribution hence promoting species survival.
- Methods of dispersal: Wind (light seeds, e.g., maize, dandelion), water (floating fruits, e.g., coconut), animals (edible fruits, sticky seeds, e.g., mango, burrs) and explosive mechanism (pods burst open, e.g., beans)
- Germination is a process where a seed develops into a seedling under suitable conditions (water, oxygen, warmth).
- Types: Epigeal (cotyledons above ground, e.g., beans) and hypogeal (cotyledons remain underground, e.g., maize)

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